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# **NPS/ 001 / 006 – Technical Specification for Insulators for Overhead Lines up to and including 132kV.**

## **1. Purpose**

The purpose of this document is to specify the technical requirements for insulators and insulator assemblies to be installed on overhead lines located in the CE Electric UK distribution network operating up to and including 132KV.

This document supersedes the following documents, all copies of which should be destroyed.

| <b>Ref</b>   | <b>Version</b> | <b>Title</b>                                                                                         |
|--------------|----------------|------------------------------------------------------------------------------------------------------|
| DSS/004/010  | 1.0            | Specification for Porcelain Insulators                                                               |
| DSS/007/010  | 1.0            | Specification for Triggered Spark Gaps                                                               |
| NEDL/MS-O/21 | 1.0            | Technical Specification for Insulators for Overhead Lines up to and Including 132kV.                 |
| NPS/001/006  | 1.0            | NPS/ 001 / 006 – Technical Specification for Insulators for Overhead Lines up to and including 132kV |

## **2. Scope**

This document applies to insulators and insulator assemblies containing porcelain, glass or composite insulators for use on overhead lines located on the CE Electric UK distribution network.

For details about the application of insulators onto the CE Electric UK distribution system see **NSP/004/127 “Guidance on the selection and application of insulators”**

The range of insulators specified within this document includes LV and service insulators rated at 650V and HV and EHV insulators rated up to and including 132kV. All Insulators shall generally be in accordance with ENA TS 43-93, Issue 4 and IEC 61109.

Insulators supplied for EHV applications will be supplied as complete assemblies including all insulator protection devices.

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## 3. Technical Requirements

### 3.1 Markings

All insulators should be marked in accordance with the requirements of BS EN 60383-1, including a system of product traceability. Markings shall be legible and indelible throughout the life of the insulator. In addition to these requirements composite string insulators shall be marked in accordance with ENA TS 43-93 or IEC 61466-1.

### 3.2 Materials

The insulators specified within this document are manufactured from a range of materials including porcelain, glass and composite materials. The associated standards are detailed below for each material type.

#### 3.2.1 Composite Materials

Composite insulators are typically manufactured from at least two insulating parts, namely a core and a housing equipped with metal end fittings. Composite insulators shall be manufactured to be fully compliant with IEC 61109 or IEC 61952 for post Insulators.

All insulator components including the polymeric materials used in the manufacture of insulators shall be traceable from the raw material supplier through the manufacturing process as required under quality assurance procedures.

The polymeric material will be described by the manufacturer in defined terms. The use of a generic term such as silicone rubber is not acceptable. Manufacturers shall complete schedule 2 with details about the material.

##### 3.2.1.1 Composite Insulator – Core

The core is the internal insulating part of a composite insulator and is designed to provide the mechanical characteristics of the insulator. To avoid brittle fracture failures, the cores shall be manufactured from an electrical grade epoxy resin matrix containing high strength, acid resistant, (low seed count) E-CR or ECR axial glass fibre reinforcement unless specified otherwise.

Post insulators that are acting purely in bending may be manufactured from E type glass as the risk of brittle fracture in these circumstances is low.

##### 3.2.1.2 Composite Insulator – Housing and sheds

The housing is the external insulating part of an insulator which provides the necessary creepage and protects the core from the weather.

Unless specified otherwise within this specification all composite insulators shall be manufactured from Silicon Rubber or EVA. Due to the variations in the make up of silicon rubber materials and the potential effects this may have on the long term performance of such a material to provide the following properties:

- UV resistance
- Tracking resistance
- Hydrophobicity

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a range of good practice characteristic values have been included within this specification

| <b>Material Composition</b>                         | <b>Minimum % per weight</b> |
|-----------------------------------------------------|-----------------------------|
| Silicone Polymer (Polydimethylsiloxane) (PDMS)      | 30%                         |
| Filler (ATH Aluminium tri hydrate and fumed silica) | 64%                         |
| Additional parts (pigments, crosslinkers)           | 1%                          |

Where manufactures differ from this characteristic composition, they shall provide supporting evidence about the long term experience of their product formulation.

The following table provides a list of characteristic properties that result from the above formulation.

Manufacturers shall provide details of their characteristic properties for consideration.  
Minimum acceptable values are detailed below.

| <b>Property</b>                                                                                                    | <b>Minimum Value</b> |
|--------------------------------------------------------------------------------------------------------------------|----------------------|
| Density                                                                                                            | 1.5g/ccm             |
| Passing Voltage level of IEC 60587                                                                                 | 4.5kV                |
| Flammability class of IEC 60695-11-10 of 3mm specimen                                                              | V0                   |
| Tensile Strength (Din 53504-S1)                                                                                    | 6 N/mm <sup>2</sup>  |
| Break Elongation (Din 53504-S1)                                                                                    | 300%                 |
| UV resistance - @ 300nm the energy of UV wave length equates to a molecular energy breakdown level of 398kJ/mole * | 445 kJ/mole          |

\* Assumed wavelength of UV light (sun) 290 – 350nm

All silicon rubber insulators shall be manufactured using the HTV (high temperature vulcanising) and shall ensure that the interface between the housing and the core is chemically bonded.

To reduce long straight axial mould lines on insulators that are manufactured using the direct moulding technique, consideration shall be given to rotating the mould line by 60° for every metre of housing length. Irrespective of the above statement, the flash or mould lines shall not exceed 1mm in height.

All rods shall be covered by a minimum insulation thickness of 3mm with a proven method for maintaining the concentricity of the silicon rubber over the rod.

### 3.2.1.3 Composite Insulator – End Fittings

End fittings transfer the mechanical load to the core. The interface between the end fitting and the housing sheath shall be sealed by an elastomer with permanent elasticity. The sealing bond shall stick to the surface of the metal fitting as well as the housing. Sealing by compression only is not considered to provide a long term sealing solution.

In the case of injection moulding technology, the sealing end fitting can be over moulded to improve the field control and to increase the available striking distance.

Preferably end fittings will be attached onto the rod by a compression method process (coaxial or hexagonal compression method), which does not damage the individual fibres of the rod. Crimp control shall be monitored by using acoustic emission devices.

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All end fittings shall be fully in accordance with IEC 60120 (Ball socket fittings). The die penetration type test within IEC 61109 shall be utilised to prove the integrity of the end fittings.

#### 3.2.1.4 Composite Insulator – Tolerances

The permitted tolerances shall meet the requirements of IEC 61109. In accordance with those standards the following tolerances are permitted for all dimensions without prior special agreement with CE Electric UK.

$\pm (0.04 \times L + 1.5)$  when  $L \leq 300\text{mm}$

$\pm (0.025 \times L + 6)$  when  $L > 300\text{mm}$ , limited to 50mm

For creepage distance, no negative tolerance is permitted.

#### 3.2.2 Ceramic Porcelain Components

Ceramic or glass used in the manufacture of insulators including low voltage line or service insulators shall comply with the requirements of IEC 60672 and ENA TS 43-93 Issue 4.

The porcelain shall be covered by a smooth and shining hard glaze free from cracks and other defects prejudicial to satisfactory performance in service. The tolerances on visual defects shall be in accordance with ENA TS 43-93 clause 4.2.1.

##### 3.2.2.1 HV Line Insulators

HV Line Insulators fitted with metal thimbles, shall be suitable for large steel heads to BS3288, Part 2 fig 2. The metal used for the thimbles shall not corrode when in contact with the cement. If lead alloy thimbles are used, the lead alloy shall have a minimum tin content of 4.8% by weight.

When specified the head of the HV porcelain insulator shall be provided with a semi-conducting glaze identifiable by its appearance.

##### 3.2.2.2 LV Line Insulators

The porcelain components of LV line insulators shall meet the requirements of ENA TS 43-93 Issue 4

#### 3.2.3 Metal Components

All ferrous materials shall be hot dip galvanising in accordance with BS EN ISO 1461 or manufactured from the appropriate grade of stainless steel.

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### 3.3 High Voltage Pin Insulators

#### 3.3.1 Pin Insulator – 11kV Earthed and Unearthed and 20kV Unearthed

Insulators shall be manufactured from porcelain with a wet withstand of 50kV. The design shall incorporate a 25mm radius groove and a 78mm diameter neck. See Appendix 2 - drawing number 1091010486 sheet 11 for details.

#### 3.3.2 Pin Insulator Anti Vandal – 11kV Earthed and Unearthed

Insulators shall be manufactured from composite material with a wet withstand of 50kV. The design shall incorporate a 25mm radius groove and a 78mm diameter neck (The Ansi standard 'F' neck insulator with a 73mm diameter may be considered as an alternative). See Appendix 2 - drawing number 1091010486 sheet 12 for details.

#### 3.3.3 Pin Insulator – 20kV Earthed and all 20kV Anti Vandal

Insulators shall be manufactured from polyethylene composite material with a wet withstand of 70kV. The design shall incorporate a 25mm radius groove and a 78mm diameter neck (See clause 3.3.2 regarding alternative option). See Appendix 2 - drawing number 1091010486 sheet 13 for details.

#### 3.3.4 Pin Insulator – 33kV Earthed and Unearthed (ENA TS 43-93 Fig. 5.2)

Insulators shall be manufactured from porcelain with a wet withstand of 90kV. They shall be supplied with a 25mm radius groove and a 120mm diameter neck.

### 3.4 Composite Post Insulators – 66kV and 132kV

Composite Line Post insulators shall be manufactured from silicon rubber materials as detailed in clause 3.2.1. Appendix 1 lists the range of insulators required and Appendix 2 provides detailed drawings. A selection of the specific creepage distance for the line post insulators shall be as detailed in clause 3.11.4

Unless specified otherwise within this document, composite post insulators shall be designed to provide a minimum cantilever load of 21kN. All tests should be carried out in accordance with IEC 61109 or IEC 61952 as appropriate.

#### 3.4.1 Conductor Clamps for Post Insulators

Unless specified otherwise within the drawing, all 66kV and 132kV line post and pilot post insulators shall be supplied complete with conductor clamps. The clamps shall be as detailed in Appendix 2 - drawing number 1091010487 sheet 6.

The clamp tops shall be designed so that:

- (i). The effect of vibration and conductor swing, both on the conductor and insulator is minimised.
- (ii). Secure clamping of the conductor is achieved without causing kinking or any undue deformation of the conductor stranding
- (iii). The clamp can accept aluminium and copper conductors.

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### 3.4.2 Composite Post Insulator Mounting Requirements

#### Flange Mounting

Unless otherwise specified, flange mounting insulators shall be provided with 4 x M16 or (5/8" – 11UFS) tapped holes on a 127mm PCD. The tapped holes shall have a threaded depth of 20mm minimum. All insulators shall be supplied complete with high tensile SAE grade 9 bolts (ISO 898-1:1999, Class 12.9) or equivalent.

#### Stud Mounting

Insulators specified as stud mounting shall be supplied with stud arrangements as specified in ENA TS 43-93 Fig 5.3.

**Note** - Due to the variation of arrangements for post insulators mounting configurations may be amended at the time of order.

### 3.4.3 Composite Post Insulator – Overvoltage protective devices

Where specified within the descriptions in Appendix 1 that the insulator shall be supplied with arcing horns. Manufactures shall provide a compatible set of overvoltage protective devices that fully comply with the requirements of clause 3.7 of this specification.

## 3.5 String Insulators

String insulators shall be used on lines with nominal system voltages up to and including 132kV. The string insulators are used in the horizontal disposition at section or terminal positions or in a vertical disposition forming part of a suspension insulator set.

Typically they are configured as detailed below

- i) 11kV – one units (unearthed supports) – (two units at earthed supports)
- ii) 20kV – two units
- iii) 33kV – three units
- iv) 66kV – five units
- v) 132kv – Horizontal disposition – 9 units
- vi) 132kV – Vertical disposition – 11 units (option to utilise 9 unit variant)

The minimum creepage distance for string insulators shall be in accordance with clause 3.11.4 of this specification.

### 3.5.1 Composite String Insulators

Composite insulators have been specified to provide insulator lengths that are multiples of the historical 140mm or 178mm centred glass or porcelain insulator strings detailed above. CE Electric UK is prepared to consider shorter insulator assemblies; however their use will be dependant upon being able to maintain adequate jumper swing clearances to tower steelwork etc. Manufactures shall quote based on compliance with existing string lengths but are encouraged to offer shorter alternatives for consideration.

The Specified Mechanical Load class and end fittings type shall be in accordance with IEC 61466-1. Ball and socket couplings shall be in accordance with BS 3288 Part 3 or IEC60120 as appropriate to the specified rating.

The insulators listed in the schedules in appendix 1 of this document that refer to "Replacement String Assemblies – 33-132kV Wood Pole & Tower Lines" shall be supplied as complete insulator assemblies (including all shackles and arcing horns as shown in the referenced drawings).

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## 3.5.2 Glass and Porcelain Discs

Glass and porcelain disc insulators shall continue to be available for specific purposes, mainly emergency repair functions on existing strings. They shall be suitable for use on high voltage overhead lines with a nominal system voltage up to and including 132kV.

These units shall be nominally 255mm diameter with 140mm or 178mm centres. Ball and socket couplings shall be in accordance with the requirements of HD 474 S1 designation 16B or 20B. The design of these interfaces shall meet the requirements of ENA TS 43-93 clause 5.1.6

Appendix 1 lists the range of insulators required with detailed drawings shown in Appendix 2.

## 3.6 Insulator Protective Devices

All 33, 66 and 132kV, composite post, pilot, tension and stay insulators shall be supplied complete with insulator protective devices (arcing gap devices). These shall be designed so as to prevent any arc current flow through the end fitting of the insulator. Manufactures shall provide arcing rings designed specifically for the insulator that attach to the hardware at each end of the tension string.

The design of the protective devices will be required to take into account the following requirements.

- Shall effectively protect the insulator units and the fittings from damage from power arcs
- Shall effectively improve the voltage distribution along the insulator string
- Shall effectively improve the corona performance of the insulator set
- Shall effectively inhibit the formation of dry band arcing on composite insulators adjacent to end fittings.

| Standard Insulator sets                                                    |                                                      |                   |                       |                                                       |
|----------------------------------------------------------------------------|------------------------------------------------------|-------------------|-----------------------|-------------------------------------------------------|
| Voltage                                                                    | Approach Set (First 1.6km)                           |                   | Normal Set            |                                                       |
|                                                                            | Non Composite Strings                                | Composite Strings | Non Composite Strings | Composite Strings                                     |
| 132 kV Tension                                                             | 1000mm (39")                                         | 1000mm            | 1120mm (23.5")        | 1175mm                                                |
| 66 kV                                                                      | 453mm (18")                                          | 440mm             | 570mm (22.5")         | 540mm                                                 |
| 33 kV                                                                      | * Not Required                                       | * Not Required    | * Not Required        | Not Required                                          |
| <b>132kV Reduced clearance suspension strings based on 9 x 140mm discs</b> |                                                      |                   |                       |                                                       |
| 132 kV                                                                     | 851mm (provides a calculated impulse value of 555kV) | 925mm             | 851mm min             | 1025mm (provides a calculated impulse value of 565kV) |

Note lab tests have shown that composite insulators have a lower electrical withstand breakdown than traditional ceramic insulators, a traditional beyond gap of 1120mm breaks down at approx 620kV and to achieve a 325kV approach gap on 66kV sets the 453mm figure needs to be reduced to 440mm

\*\*\* It is assumed that surge arrestors will always be installed on 33kV Cable terminations

Where 275kV or 400kV lines are operated at 132kV, extended arcing horns shall be used for the first 1.6km from substations or sealing ends to obtain a 1000mm arc gap.

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## 3.7 Stay Insulators

Stay insulators shall be manufactured in accordance with the electrical and mechanical loadings detailed in ENA TS 43-91.

### LV – 33kV Stay Insulators

Shall be manufactured from porcelain and designed to comply with the dimensions of type 1 and type 2 arrangements as shown in figure 4.3 of ENA TS 43-91.

### 66 – 132kV Stay Insulators (earthed assemblies)

Shall be manufactured from porcelain and be designed to be in compliance with ENA TS 43-91 type 2.

### 66 – 132kV Stay Insulators (unearthed assemblies)

Shall be single composite insulator assemblies complete with overvoltage protection arcing horns that comply with clause 3.6. The assemblies shall be designed to comply with ENA TS 41-91, IEC 61109 and IEC 61466. See drawing 1091010372 sht 2 for more details.

## 3.8 Low Voltage Line and Service Insulators

### 3.8.1 Reel Insulator

Reel insulators for general use on low voltage distribution networks and service arrangements shall be in accordance with ENA TS 43-93 Section 9, with dimensions to Fig. 9.1.

### 3.8.2 Coach Screw Service Insulator

Coach screw insulators for general use on service arrangements shall be in accordance with ENA TS 43-93 Section 9, with dimensions to Fig. 9.2.

## 3.9 Triggered Spark Gap

Triggered spark gaps are a porcelain insulator assembly used to provide over voltage protection on 11, 20 and 33kV wood pole overhead lines. The unit shall be supplied with two sets of arcing horns to facilitate an early flashover of high over voltages. The arc gap dimensions are detailed below. Designs using composite material may be considered providing manufacturers can provide evidence of composite materials working in similar working conditions

| Voltage (kV) | Triggered Gap (mm) | Main Arc Gap (mm) | Drawing Number        |
|--------------|--------------------|-------------------|-----------------------|
| 11           | 3                  | 25                | 1.09.101.0603 Sheet 1 |
| 20           | 10                 | 35 - 38           | 1.09.101.0603 Sheet 2 |
| 33           | 10                 | 45 - 48           | 1.09.101.0603 Sheet 2 |

Appendix 1 lists the range of spark gaps required with detailed drawings shown in Appendix 2.

## 3.10 General Design Requirements for Insulators

### 3.10.1 Testing Requirements

Design, type, sample, and routine tests shall be as specified in ENA TS 43-93 Clauses 6.2, 6.3 and 6.4.

|                           |                |                        |          |             |    |           |    |
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### 3.10.2 Tracking and Erosion Tests

Shall be as detailed in ENA TS 43-93 clause 6.5 and shall include the following:

- 1000 hour salt fog test as defined in IEC 61109 clause 5.3.2
- 5000 hour multi-stress weather aging test as defined in IEC 61109 Annex C.
- A test to IEC 60587-2007 using either the method 1 or 2

### 3.10.3 Electrical withstand voltages

The design of the insulators shall be such that the electrical withstand voltages below are achieved. Tests shall be performed in accordance with BS EN 60383-1 for insulator units, line post insulators and pin insulators, BSEN 60383-2 for insulator strings and sets, IEC 61109 for composite string insulators and IEC 61952 for composite line post insulators.

| <b>Nominal System Voltage (kV)</b>               | <b>11</b> | <b>20</b> | <b>33</b> | <b>66</b> | <b>132</b> |
|--------------------------------------------------|-----------|-----------|-----------|-----------|------------|
| Wet 1 min power frequency withstand voltage (kV) | 45        | 70        | 90        | 140       | 275        |
| Dry impulse withstand voltage (kV)               | 95        | 125       | 200       | 325       | 650        |

### 3.10.4 RIV Requirements and Corona Extinction Voltage

Shall be as detailed in ENA TS 43-93 clause 6.6

### 3.10.5 Pollution Levels and Creepage Distances

Unless specified otherwise the minimum creepage distances for high voltage insulators and insulator sets shall be as defined in the following table extracted from ENA TS 43-93 Table 7.1.

| <b>System Voltage (kV)</b>       | <b>Pollution level</b>              | <b>11</b> | <b>20</b> | <b>33</b> | <b>66</b> | <b>132</b> |
|----------------------------------|-------------------------------------|-----------|-----------|-----------|-----------|------------|
| Vertical (Pin, post, suspension) | <b>Medium Class II</b><br>(20mm/kv) | 220 mm    | 440mm     | 660mm     | -         | -          |
| Horizontal (Tension)             |                                     | 154 mm    | 308mm     | 462mm     | -         | -          |
| Vertical (Pin, post, suspension) | <b>Heavy Class III</b><br>(25mm/kV) | -         | -         | -         | 1650mm    | 3300mm     |
| Horizontal (Tension)             |                                     | -         | -         | -         | 1156mm    | 2310mm     |

## 3.11 Mechanical Performance

### 3.11.1 Porcelain or Glass Insulators

The required mechanical rating of glass or porcelain insulators shall be in accordance with the requirements of ENA TS 43-93 clause 8.1

### 3.11.2 Composite Insulators

The required mechanical rating of composite insulators shall be in accordance with the requirements of ENA TS 43-93 clause 8.2. In summary suspension insulators shall be rated at 70kN and tension insulators shall be rated at a minimum of 120 or 190kN.

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## 4. References

### 4.1 External Documentation

| Reference                                       | Title                                                                                                                                                                                                         |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENA TS 43-93 – 4: 2004<br>BS EN 60383 – 1: 1998 | Technical Specification for Line Insulators<br>Insulators for lines with a nominal voltage above 1000v: Ceramic or glass insulator units for a.c. systems – Definitions, test methods and acceptance criteria |
| IEC 61466 – 1: 1997                             | Composite string insulator units for overhead lines with a nominal voltage greater than 1000v: Standard strength classes and end fittings                                                                     |
| IEC 60672: 1995<br>BS 3288 – 2: 1990            | Ceramic and glass insulating materials<br>Insulator and conductor fittings for overhead power lines: Specification for a range of fittings.                                                                   |
| IEC 61109: 1995                                 | Composite Insulators for Overhead Power Lines With a Nominal Voltage Greater than 1000v                                                                                                                       |
| BS EN ISO 1461: 1999                            | Hot dipped galvanised coatings on fabricated iron and steel articles – specifications and test methods                                                                                                        |
| IEC 61466 – 1: 1997                             | Composite string insulator units for overhead lines with a nominal voltage greater than 1000v: Standard strength classes and end fittings                                                                     |
| IEC 60815: 1986<br>ENA TS 41-91 – 5: 2004       | Guide for the selection of insulator in polluted conditions<br>Technical Specification for Stay Strands and Stay Fittings for Overhead Lines                                                                  |
| IEC 60437: 1997<br>IEC 60507: 1991              | Radio interference test on high voltage insulators<br>Artificial pollution tests on high voltage insulators to be used on a.c. systems.                                                                       |
| IEC 60587:2007                                  | Test methods for evaluating resistance to tracking and erosion                                                                                                                                                |

### 4.2 Internal documentation

| Reference   | Title                                                         |
|-------------|---------------------------------------------------------------|
| NPS/004/127 | Guidance note on the selection and application of insulators. |

### 4.3 Summary of Changes

- Clause 3.6 – amended to reflect the changes in arc gap requirements on arc protective devices were composite insulators are used.
- Appendix 1 – Descriptions and catalogue numbers amended to reflect the introduction of composite insulators and the additional variations created by the concept of approach and beyond arc protection gaps on the insulator assemblies.

|                           |     |                        |          |             |    |           |    |  |
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## 5. Definitions

| <b>Term</b>      | <b>Definition</b>                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brittle fracture | Brittle fracture is a failure mode where stress-corrosion cracking (SCC) of glass-reinforced polymer (GRP) can result in total mechanical failure of the insulator |
| Type ECR or E-CR | Electrolytic Corrosion Resistant (Low or boron free glass )                                                                                                        |
| Seed Count       | Seeds are gaseous inclusions (voids) inside the fibres left from the glass fibre manufacturing process                                                             |
| Type E           | Electrical Grade Glass                                                                                                                                             |
| Earthed          | Earthed Assemblies are wood poles or steel structures where the crossarm steelwork is electrically connected to earth and hence provides a lower BIL               |
| Unearthed        | Unearthed structures generally provide a higher BIL as they are insulated from earth                                                                               |
| BIL              | Basic Insulation Level                                                                                                                                             |
| EVA material     | Is a co-polymer of Ethylene and Ethylene Vinyl Acetate                                                                                                             |

|                    |                |                 |          |      |    |    |    |
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## 6. Authority for issue

### 6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

|          |                                  | Sign | Date |
|----------|----------------------------------|------|------|
| G Hammel | Overhead System Standards Manage |      |      |

### 6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|             |                         | Sign | Date |
|-------------|-------------------------|------|------|
| M Nicholson | Head of System Strategy |      |      |

### 6.3 Technical Assurance

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### 6.4 DBD Assurance

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|-----------|-------------------|------|------|
| S Johnson | DBD Administrator |      |      |

### 6.5 Approval

Approval is given for the content of this document

|              |                                   | Sign | Date |
|--------------|-----------------------------------|------|------|
| C Holdsworth | Standards and Assessments Manager |      |      |

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## Appendix 1 – Schedule of Items

### Pin Insulators – (11- 33kV Wood Pole Lines)

| Cat. No. | Description                                                                                                                                | Drawing<br>(CE Electric or ENA TS)            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 253777   | Insulator: 11/20kV Brown Porcelain Pin: 50kV Wet Withstand, 78mm Neck & 25mm Groove                                                        | 1091010486 sht. 11                            |
| 253701   | Insulator: 11kV Grey Composite Anti-Vandal Pin: 50kV Wet Withstand 73mm Neck & 25mm Groove                                                 | 1091010486 sht. 12                            |
| 253833   | Insulator: 20kV Grey Composite Pin Suitable For 20kV Earth Applications Including Anti-Vandal: 55kV Wet Withstand, 73mm Neck & 25mm Groove | 1091010486 sht. 13                            |
| 253423   | 33kV Brown Porcelain Pin: 90kV Wet Withstand, 120mm Neck & 25mm Groove                                                                     | ENA TS 43-93 Fig. 5.2 or 109.101.00486 Sht. 4 |

### Composite Post Insulators – (66 – 132kV Wood Pole Lines)

| Cat. No | Description                                                                                                                                                                        | CE-Electric<br>Drawing                   |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 253703  | Insulator: 66kV Grey Composite Line Post With Flange Mounting (8 x 18mm holes spaced on 225mm PCD) Replacement For OHL4 Porcelain Post.                                            | 1091010589 Sht.13                        |
| 253702  | Insulator: 66kV Grey Composite Line Post Complete With Single 50mm Mounting Stud                                                                                                   | 1091010487 Sht.12                        |
| 253704  | Insulator: 66kV Grey Composite Station Post: For Cable Support Applications.                                                                                                       | 1091010487 Sht.28                        |
| 251551  | 132kV Polymeric Vertical Post Insulators c/w with Clamp Adaptor and Trunnion designed for poplar conductor.                                                                        | 1091010487 sht. 18                       |
| 251550  | 132kV Polymeric Pilot Post Insulators c/w Armour Grip Clamps designed for poplar conductor, arcing horn protection & support stool to increase the overall height of the assembly. | 1091010487 sht.18,<br>1091010487 sht. 34 |
| 251548  | 132kV Polymeric Horizontal Post Insulator c/w with Clamp Top and Trunnion clamp designed for use with poplar conductor (no bendable base)                                          | 1091010487 sht.16                        |
| 251549  | 132kV Polymeric Horizontal Post Insulator c/w OHL9 bendable flat base, Clamp Top & Trunnion clamp designed for poplar conductor and use on OHL 9 construction                      | 1091010487 sht.16                        |

|                           |     |                        |          |             |    |           |    |
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**Composite Post Insulators (66 – 132kV Wood Pole Lines) (cont)**

| <b>Cat. No</b> | <b>Description</b>                                                                                                                                                                                           | <b>CE-Electric Drawing</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 346397         | Arcing Horn Kit and extension Arc Horn for application onto item insulator drawing (1091010487 sht. 18) Vertical Line Post Insulators in approach zones only (Excludes Insulator) as per arrangement drawing | 1091010487 sht.32          |
| 346398         | Arcing Horn Kit and extension Arc Horn for application onto insulator drawing (1091010487 sht.16) Horizontal Line Post Insulators in approach zones only (Excludes Insulator) as per arrangement drawing     | 1091010487 sht.33          |

**Composite String Insulators – (11 – 33kV Wood Pole Lines)**

| <b>Cat. No</b> | <b>Description</b>                                                                                                                                                                                           | <b>Drawing<br/>(CE Electric or ENA TS)</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 253695         | 11kV earthed and unearthed construction and 20kV unearthed construction 70kN Composite Tension Insulator: 50kV Wet Withstand, 95kV BIL (2 x 140mm spacing) 16mm Ball/Socket End Fitting                      | 1091010487 sht. 22                         |
| 216135         | 11kV - 70kN Composite Flying Section Insulator, 45kV Wet withstand, 95kV BIL (2 x 140mm spacing) 16mm Ball / Ball End Fitting                                                                                | 1091010485 sht. 10                         |
| 253706         | 20kV (Earthed construction) - 70kN Composite Tension Insulator: 70kV Wet Withstand, 125kV BIL (2 x 140mm spacing) 16mm Ball/Socket End Fitting.                                                              | 1091010487 sht. 24                         |
| 247412         | 20kV - 70kN Composite Flying Section Insulator, 70kV Wet withstand, 125kV BIL (3 x 140mm spacing): 16mm Ball / Ball End Fitting.                                                                             | 1091010485 sht. 11                         |
| 216150         | 33kV Insulator (43-40 Tension & CE/C37 Suspension Insulator) Earthed or unearthed assembly 70kN Composite Tension/Suspension: 90kV Wet Withstand, 200kV BIL (3 x 140mm spacing) 16mm Ball/Socket End Fitting | 1091010487 sht. 23                         |
| 251546         | 33kV (CE/C37 Tension Insulator Earthed and Unearthed Insulator construction) 120kN Composite Tension: 90kV Wet Withstand, 200kV BIL (3 x 178mm spacing) 20mm Ball/Socket End Fitting                         | TBA                                        |

|                           |     |                        |          |             |    |           |    |
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**Composite String Insulators – (66 – 132kV Woodhouse Mast replacement and OHL9 & OHL10 Specification)**

| <b>Cat. No</b> | <b>System Voltage (kV)</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                | <b>Drawing</b>                           |
|----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 251553         | 132kV                      | 132kV Tension Set Assembly (Approach)<br>To be supplied complete with 120kN insulators with ball socket couplings to IEC 60120/20 and all fittings from Earth End Shackle to Live End socket tongue, including arcing horns (1000mm gap) but excluding compression terminations as shown on manufactures drawing.                                                                 | 1091010487 sht. 29<br>1091010487 sht. 31 |
| 251554         | 132kV                      | 132kV Tension Set Assembly (Approach) – 66kV Operation<br>Supplied complete with 120kN insulators with ball socket couplings to IEC 60120/20 and all fittings from Earth End Shackle to Live End socket tongue, including arcing horns & Arc horn extensions (440mm gap) for 66kV but excluding compression terminations                                                          | 1091010487 sht. 29<br>1091010487 sht. 31 |
| 251552         | 33,66 & 132kV              | 132kV Tension Set Assembly (Non Approach) – used for 33, 66 or 132kv Woodhouse tension situations.<br>Supplied complete with 120kN insulators with ball socket couplings to IEC 60120/20 and all fittings from Earth End Shackle to Live End socket tongue, including arcing horns (adjustable gap from 1175mm at 132kV to 540mm at 66kV) but excluding compression terminations. | 1091010487 sht. 29<br>1091010487 sht. 30 |

|                           |                |                        |          |             |    |           |    |
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### Replacement Insulator String Assemblies – (33-132kV Wood Pole and Tower Lines)

| <b>66kV Wood Pole Lines – Suspension Structures (Excluding OHL9 &amp; OHL10 constructions)</b> |             |                                                                                                                                                                                                                                               |                                        |
|------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Cat Number</b>                                                                              | <b>Item</b> | <b>Purpose for Insulator assembly</b>                                                                                                                                                                                                         | <b>Existing drawing reference</b>      |
| 251542                                                                                         | 1           | 66kV Suspension assembly for OHL4 or CE/C/37 construction ( 70kN) based on 5 x 140mm spacing, 140kV Wet Withstand , 325kV BIL, 1650mm creep - 16mm ball hook fitting and socket clevis – Approach arc gap 440mm.                              | Composite version of 1091010488 sht. 6 |
| 251541                                                                                         | 1a          | 66kV Suspension assembly for OHL4 or CE/C/37 construction (70kN) based on 5 x 140mm spacing, 140kV Wet Withstand , 325kV BIL, 1650mm min creep - 16mm ball hook fitting and socket clevis - Unearthed or Normal construction (540mm arc gap). | Composite version of 1091010488 sht. 6 |
| <b>66kV Wood Pole Lines – Tension Structures (Excluding OHL9 &amp; OHL10 constructions)</b>    |             |                                                                                                                                                                                                                                               |                                        |
| 251539                                                                                         | 2           | 66kV Tension Assembly for CE/C37 construction (120kN), based on 5 x 178mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 20mm ball ended eye link fitting and socket tongue, approach arc gap = 440mm                            | Composite version of 1091010428 sht. 7 |
| 251536                                                                                         | 2a          | 66kV Tension Assembly for CE/C37 construction (120kN), based on 5 x 178mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 20mm ball ended eye link fitting and socket tongue, 540mm gap on unearthed or normal construction       | Composite version of 1091010428 sht. 7 |
| 251559                                                                                         | 3           | 66kV Tension Assembly for OHL4 construction (70kN), based on 5 x 140mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 16mm ball ended eye link fitting and socket tongue, approach arc gap = 440mm                               | Composite version of 1091010488 sht. 2 |
| 251560                                                                                         | 3a          | 66kV Tension Assembly for OHL4 construction (70kN), based on 5 x 140mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 16mm ball ended eye link fitting and socket tongue, 540mm gap on unearthed or normal construction          | Composite version of 1091010488 sht. 2 |

|                           |     |                        |          |             |    |           |    |
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### Replacement Insulator String Assemblies – (33-132kV Wood Pole and Tower Lines) Continued

| 66-132kV Tower or mast lines – Suspension structures |      |                                                                                                                                                                                                                                                 |                                           |
|------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Cat Number                                           | Item | Purpose for Insulator assembly                                                                                                                                                                                                                  | Existing drawing reference                |
| 251535                                               | 4    | 66kV Suspension Assembly (Tower Lines) 70kN, 5 x 140mm spacing, 140kV Wet Withstand, 325kV BIL, 1650mm min creep, with 16mm ball hook (15/32) and socket clevis fittings - 440mm arcing horn gap on approach structures                         | Composite equivalent of 1091010428 sht. 7 |
| 251534                                               | 4a   | 66kV Suspension Assembly (Tower Lines) 70kN, 5 x 140mm spacing, 140kV Wet Withstand, 325kV BIL, 1650mm min creep with 16mm ball ended hook (15/32) and socket clevis fittings - 540mm arcing horn gap on Non approach structures                | Composite equivalent of 1091010428 sht. 7 |
| 251533                                               | 5    | 132kV Suspension Assembly (Tower Lines) 70kN, 11 x 140mm spacing, 275kV Wet Withstand, 650kV BIL, 3300mm min creep with 16mm ball ended hook (15/32) and socket clevis fittings - 1000mm arcing horn gap on approach construction               | Composite equivalent of 1091010428 sht. 7 |
| 251561                                               | 5a   | 132kV Suspension Assembly (Tower Lines) 70kN, 11 x 140mm spacing, 275kV Wet Withstand, 650kV BIL, 3300mm min creep with 16mm ball ended hook (15/32) and socket clevis fittings - 1175mm arcing horn gap on non approach or normal construction | Composite equivalent of 1091010428 sht. 7 |
| 251562                                               | 5b   | 132kV Suspension Assembly - <b>reduced clearance</b> (Tower Lines) 70kN, 9 x 140mm spacing with 16mm ball ended hook (15/32) and socket clevis fittings - 925mm arcing gap for approach construction.                                           | Composite equivalent of 1091010428 sht. 7 |
| 251531                                               | 5c   | 132kV Suspension Assembly - <b>reduced clearance</b> (Tower Lines) 70kN, 9 x 140mm spacing with 16mm ball ended hook (15/32) and socket clevis fittings – 1025mm arcing gap for non approach or normal construction                             | Composite equivalent of 1091010428 sht. 7 |

|                           |                |                        |          |             |    |           |    |
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### Replacement Insulator String Assemblies – (33-132kV Wood Pole and Tower Lines) Continued

| <b>66-132kV Tower or mast lines – Tension structures</b>             |             |                                                                                                                                                                                                                                   |                                                           |
|----------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Cat Number</b>                                                    | <b>Item</b> | <b>Purpose for Insulator assembly</b>                                                                                                                                                                                             | <b>Existing drawing reference</b>                         |
| 251530                                                               | 6           | 66kV Tension Assembly for Tower Lines, (120kN) based on 5 x 178mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 20mm ball ended eye link fitting and socket tongue, 440mm arc gap on approach construction          | Composite equivalent of 1091010428 sht. 8<br>66kV version |
| 251563                                                               | 6a          | 66kV Tension Assembly for Tower Lines, (120kN) based on 5 x 178mm spacing, 140kV Wet Withstand , 325kV BIL, 1156mm min creep, 20mm ball ended eye link fitting and socket tongue, 540mm arc gap on non approach construction      | Composite equivalent of 1091010428 sht. 8<br>66kV version |
| 251528                                                               | 7           | 132kV Tension Assembly for Tower Lines, (120kN) based on 9 x 178mm spacing, 275kV Wet Withstand, 650kV BIL, 2310mm min creep, 20mm ball ended eye link fitting and socket tongue with 1000mm arc gap on approach construction     | Composite equivalent of 1091010428 sht. 8                 |
| 251564                                                               | 7a          | 132kV Tension Assembly for Tower Lines, (120kN) based on 9 x 178mm spacing, 275kV Wet Withstand, 650kV BIL, 2310mm min creep, 20mm ball ended eye link fitting and socket tongue with 1175mm arc gap on non approach construction | Composite equivalent of 1091010428 sht. 8                 |
| <b>66-132kV Tower or mast lines – Low Duty Down lead Structures)</b> |             |                                                                                                                                                                                                                                   |                                                           |
| 251626                                                               | 8           | 66kV Low Duty Assembly for down leads to include for both upright and inverted sets - 70kN, 2 x (5 x 140mm spacing) with 16mm ball ended eye link and socket tongue - 440mm arcing horn gap                                       | Composite equivalent of 1091010428 sht9<br>66kV version   |
| 251525                                                               | 9           | 132kV Low Duty Assembly for down leads to include both upright and inverted sets - 70kN, 2 x (11 x 140mm spacing) with 16mm ball ended eye link and socket tongue fittings - 1000mm arcing horn gap                               | Composite equivalent of 1091010428 sht9                   |

|                           |                |                        |          |             |    |           |    |
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### Replacement Insulator String Assemblies – (33-132kV Wood Pole and Tower Lines) Continued

| 132kV Tower Lines – L3 / L7 Construction |             |                                                                                                                                                                                            |                                         |
|------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Cat Number</b>                        | <b>Item</b> | <b>Purpose for Insulator assembly</b>                                                                                                                                                      | <b>Existing drawing reference</b>       |
| 251524                                   | 10          | 132kV Tension Assembly – (Twin Lynx Tower Lines) - (120kN) 20mm ball ended eye link fitting and socket tongue based on 2 off 9 x 178mm spacing assemblies with 1000mm gap on approach      | Composite version of 1091010428 sht10   |
| 251523                                   | 10a         | 132kV Tension Assembly – (Twin Lynx Tower Lines) - (120kN) 20mm ball ended eye link fitting and socket tongue based on 2 off 9 x 178mm spacing assemblies with 1175mm gap on non-approach. | Composite version of 1091010428 sht10   |
| 251522                                   | 11          | 132kV Suspension (125kN) Assembly – (Twin Lynx or single large conductor Tower Line), 11 x 171mm spacing with 1000mm arc gap on approach.                                                  | composite version of 1091010428 sht. 11 |
| 251521                                   | 11a         | 132kV Suspension (125kN) Assembly – (Twin Lynx or single large conductor Tower Line), 11 x 171mm spacing with 1175mm arc gap on Non Approach.                                              | composite version of 1091010428 sht. 11 |
| 251520                                   | 12          | 132kV Low Duty Assembly for 120kN down leads - To include both Upright and Inverted insulator assemblies, 1000mm arc gap.                                                                  | composite version of 1091010428 sht. 12 |
| 251519                                   | 13          | 132kV Tension Assembly – (Single large Conductor 190kN) based on 9 x 178mm spacing with 1000mm gap on approach construction                                                                | Composite version of 1091010428 sht13   |
| 251565                                   | 13a         | 132kV Tension Assembly – (Single large Conductor 190kN) based on 9 x 178mm spacing with 1175mm gap on non approach construction.                                                           | Composite version of 1091010428 sht13   |

|                           |     |                        |          |             |    |           |    |
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**Glass and Porcelain Discs – as detailed in ENA TS 43-93 Fig. 5.5**

| <b>Cat Number</b> | <b>Purpose for Insulator assembly</b>                                                                                 | <b>CE Electric Drawing Reference</b>         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 250529            | Glass Disk 254mm x 140mm, 70kN MFL with 16mm Ball/Socket coupling                                                     | 1.09101.0484 Sht. 2 and ENA TS 43-95 fig 5.5 |
| 253584            | Grey Porcelain Tension Disc 292mm x 178mm, 125kN MFL with 20mm Ball/Socket coupling                                   | 109.101.0485 sht. 8                          |
| 253616            | Porcelain Disc 285mm x 140mm, 70kN MFL, Standard, Anti-Fog suspension with 16mm Ball/Socket coupling (66/132kv lines) | 109.101.0485 sht. 4                          |
| 253438            | Glass Disc 280mm x 178mm, 125kN MFL, with 20mm Ball/Socket couplings (66/132kv lines)                                 | 109.101.0485 sht. 2                          |
| 253476            | Glass (Anti-Fog) Disc 254 x 140mm, 70kN MFL, with 16mm Ball/Socket Couplings                                          | 1091010485 sht. 3                            |
| 216911            | Brown Porcelain Disc 318mm x 171mm, 125kN MFL, Low Duty suspension disc with 20mm Ball/Socket Couplings               | Y707L0707                                    |
| 216820            | Glass Tension Disc 305mm x 200mm, 190kN MFL, (132kV Zebra or Rubas lines)                                             | Y707L0703                                    |

**Stay Insulators – All Voltages**

| <b>Cat. No</b> | <b>System Voltage (kV)</b> | <b>Description</b>                                                                                                                                                                                  | <b>Drawing (CE Electric or ENA TS)</b>                  |
|----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 248232         | 0.24 to 20                 | Stay Insulator Porcelain designed in accordance with ENA TS 43-90 Type 1 (Brown)                                                                                                                    | ENA TS 43-93 Fig. 4.3 and 1.00.043.9107 Sheet 1-Item 1. |
| 253743         | 33 to 66                   | Stay Insulator Porcelain designed in accordance with ENA TS 43-90 Type 2 (Brown)                                                                                                                    | ENA TS 43-93 Fig. 4.3 and 1.00.043.9107 Sheet 1-Item 2  |
| 251547         | 66 and 132                 | 132kV SINGLE Stay Insulator Assembly (unearthed) with 120kN ball socket couplings all fittings from the earthed end 90° Shackle to the live end shackle including arcing horns as shown on drawing. | 109.101.0372 sheet 2                                    |

|                           |     |                        |          |             |    |           |    |
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## Low Voltage Line and Service Insulators

| <b>Cat. No</b> | <b>Description</b>                                         | <b>Drawing (CE Electric or ENA TS)</b>    |
|----------------|------------------------------------------------------------|-------------------------------------------|
| 253917         | Low Voltage Reel Insulator, Brown Porcelain                | 1.00.043.9304 or<br>ENA TS 43-93 Fig. 9.1 |
| 253866         | Low Voltage Coach Screw Service Insulator, Brown Porcelain | 1.00.043.9305 or<br>ENA TS 43-93 Fig. 9.2 |

## Triggered Spark Gap

| <b>Cat. No</b> | <b>System Voltage (kV)</b> | <b>Triggered Gap (mm)</b> | <b>Main Arc Gap (mm)</b> | <b>Drawing (CE Electric or ENA TS)</b> |
|----------------|----------------------------|---------------------------|--------------------------|----------------------------------------|
| 242471         | 11                         | 3                         | 25                       | 1.09.101.0603 Sht. 1                   |
| 242537         | 20 and 33                  | 10                        | 35 - 48                  | 1.09.101.0603 Sht. 2                   |

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Feb - 10 | <b>Page</b> | 23 | <b>of</b> | 49 |

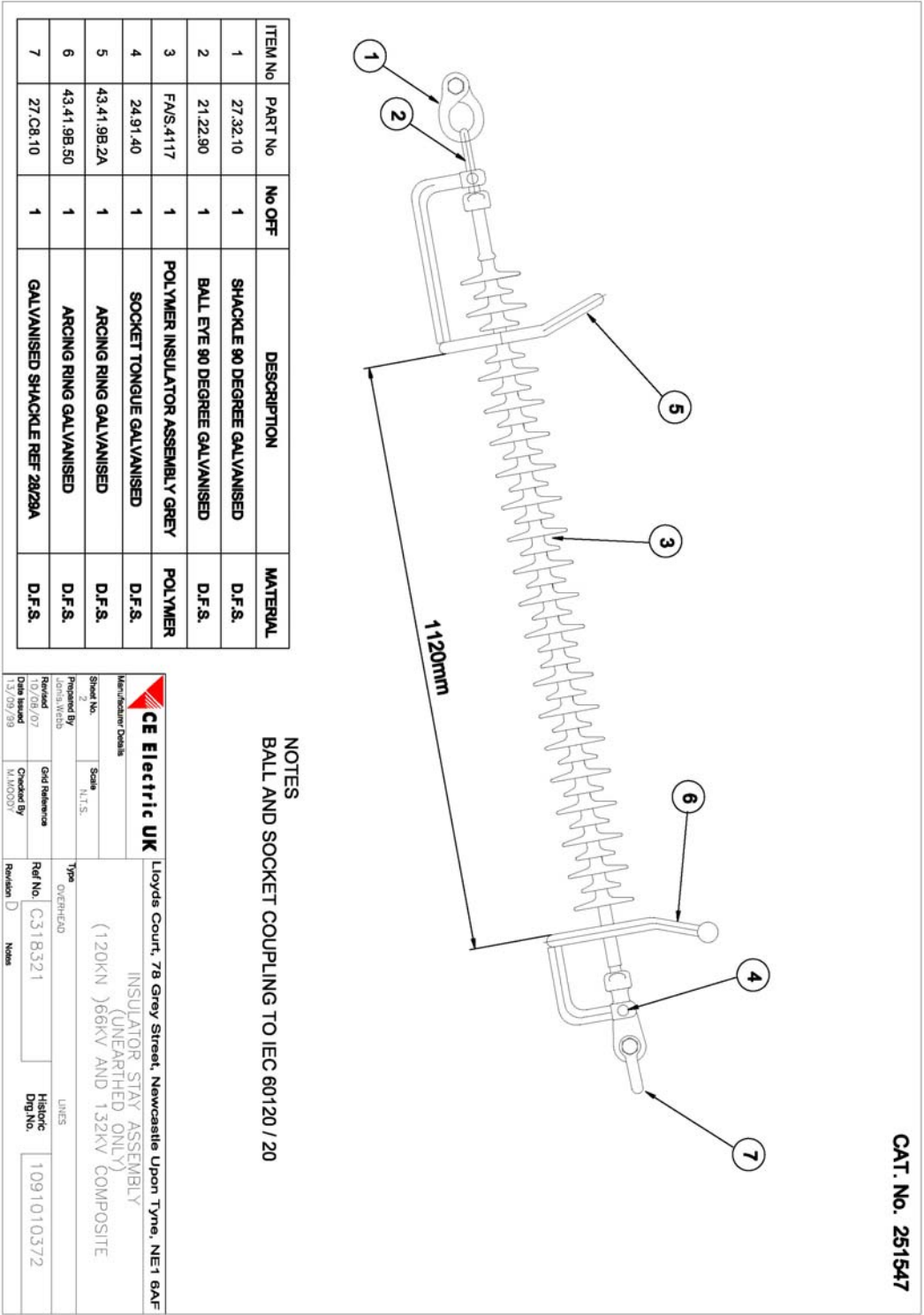
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## Appendix 2 - Material Composition

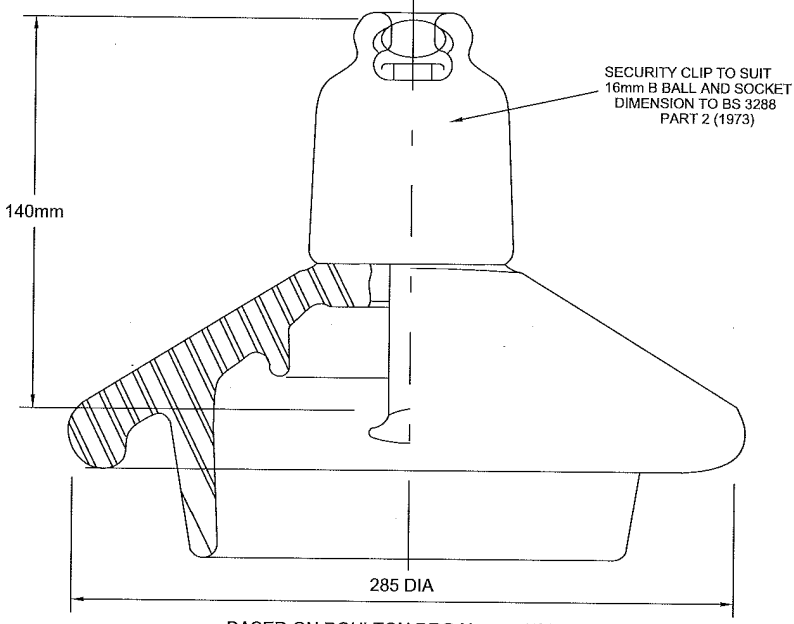
| Material Composition | Minimum % per weight | Supplier of Materials |
|----------------------|----------------------|-----------------------|
| Silicon Polymer      |                      |                       |
| Filler               |                      |                       |
| Additional Pigments  |                      |                       |
| Type of Glass Rod    |                      |                       |

**\*\*\* Appendix 2 must be completed by all Suppliers \*\***

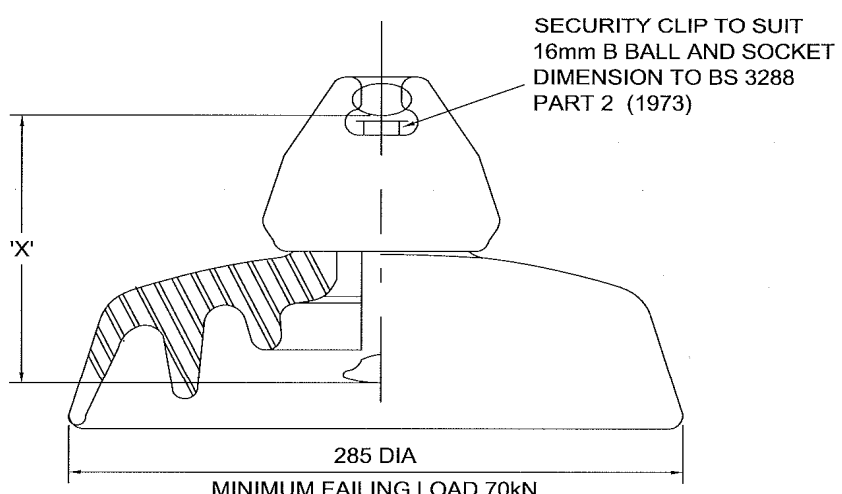
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| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 24 of 49 |



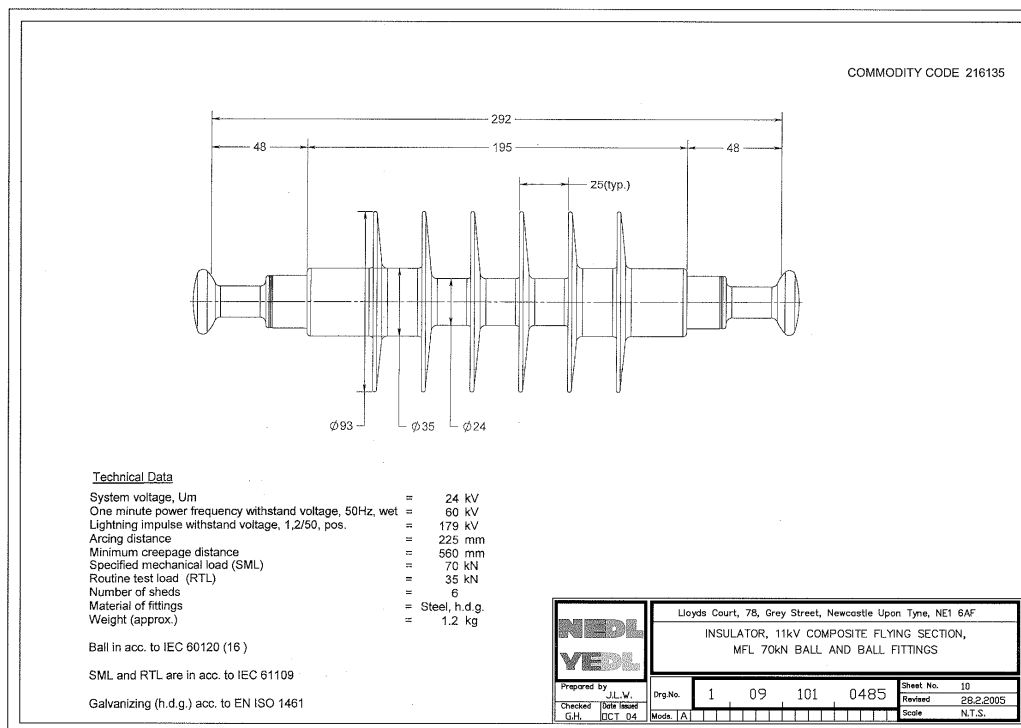
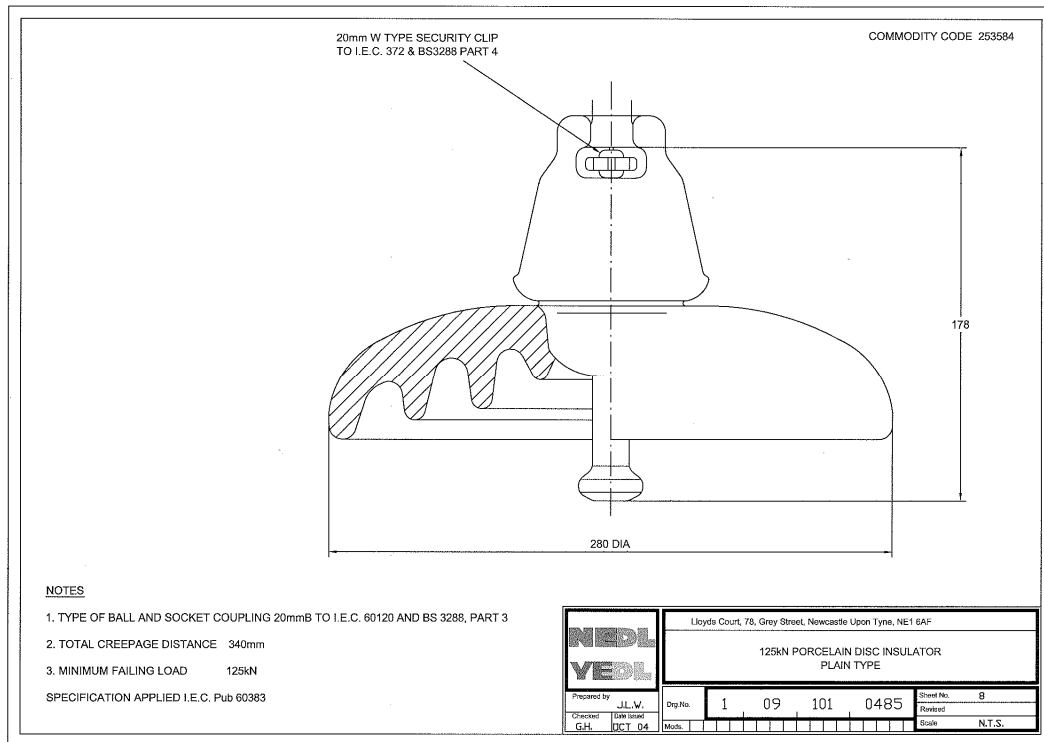
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|--------------------|----------------|-----------------|----------|------|----|----|----|
| Document reference | NPS/ 001 / 006 |                 |          |      |    |    |    |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 25 | of | 49 |

|                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|-----------|------|-----|-----------------|--|--|--------------|
| NORTHERN ELECTRIC CAT No 253476                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
|  <p style="text-align: center;">140mm</p> <p style="text-align: center;">285 DIA</p> <p style="text-align: center;">BASED ON DOULTON DRG Nos. s - 70306</p>                                                                      |                                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| <p><b>NOTE</b><br/>FOR USE IN SUSPENSION SETS ON NORMAL 66KV AND 132KV OVERHEAD LINES<br/>APPROVED TYPES TO BS 137, PART 2:- DIM X = 140</p>                                                                                                                                                                        |                                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| ALL DIMENSIONS SHOWN ARE IN MILLIMETRES UNLESS OTHERWISE STATED                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| <div style="border: 2px solid black; padding: 5px; text-align: center; font-weight: bold; font-size: 1.2em;">NEDL<br/>YEDL</div>                                                                                                                                                                                    | <p style="text-align: center;">Lloyds Court, 78, Grey Street, Newcastle Upon Tyne NE1 6AF</p> <p style="text-align: center; font-weight: bold;">STRING INSULATOR UNIT</p> <p style="text-align: center;">GLASS (MFL 70kN)</p> |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%;">Prepared By<br/>J.L.W.</td> <td style="width: 50%;">Sheet No. 3</td> </tr> <tr> <td>Checked<br/>G.H.</td> <td>Revised. 2.3.05</td> </tr> <tr> <td>Date Issued<br/>APR '79</td> <td>Scale N.T.S.</td> </tr> </table> | Prepared By<br>J.L.W.                                                                                                                                                                                                         | Sheet No. 3     | Checked<br>G.H. | Revised. 2.3.05 | Date Issued<br>APR '79 | Scale N.T.S. | <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 20%;">Drg.No.</td> <td style="width: 40%;">1 09 101 0485</td> <td style="width: 40%;">Drg.No. 3</td> </tr> <tr> <td>Mod.</td> <td>F G</td> <td>Revised. 2.3.05</td> </tr> <tr> <td colspan="2"></td> <td>Scale N.T.S.</td> </tr> </table> | Drg.No. | 1 09 101 0485 | Drg.No. 3 | Mod. | F G | Revised. 2.3.05 |  |  | Scale N.T.S. |
| Prepared By<br>J.L.W.                                                                                                                                                                                                                                                                                               | Sheet No. 3                                                                                                                                                                                                                   |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| Checked<br>G.H.                                                                                                                                                                                                                                                                                                     | Revised. 2.3.05                                                                                                                                                                                                               |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| Date Issued<br>APR '79                                                                                                                                                                                                                                                                                              | Scale N.T.S.                                                                                                                                                                                                                  |                 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| Drg.No.                                                                                                                                                                                                                                                                                                             | 1 09 101 0485                                                                                                                                                                                                                 | Drg.No. 3       |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
| Mod.                                                                                                                                                                                                                                                                                                                | F G                                                                                                                                                                                                                           | Revised. 2.3.05 |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |
|                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                               | Scale N.T.S.    |                 |                 |                        |              |                                                                                                                                                                                                                                                                                                                                     |         |               |           |      |     |                 |  |  |              |

|                    |                |                 |          |      |    |    |    |
|--------------------|----------------|-----------------|----------|------|----|----|----|
| Document reference | NPS/ 001 / 006 |                 |          |      |    |    |    |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 26 | of | 49 |

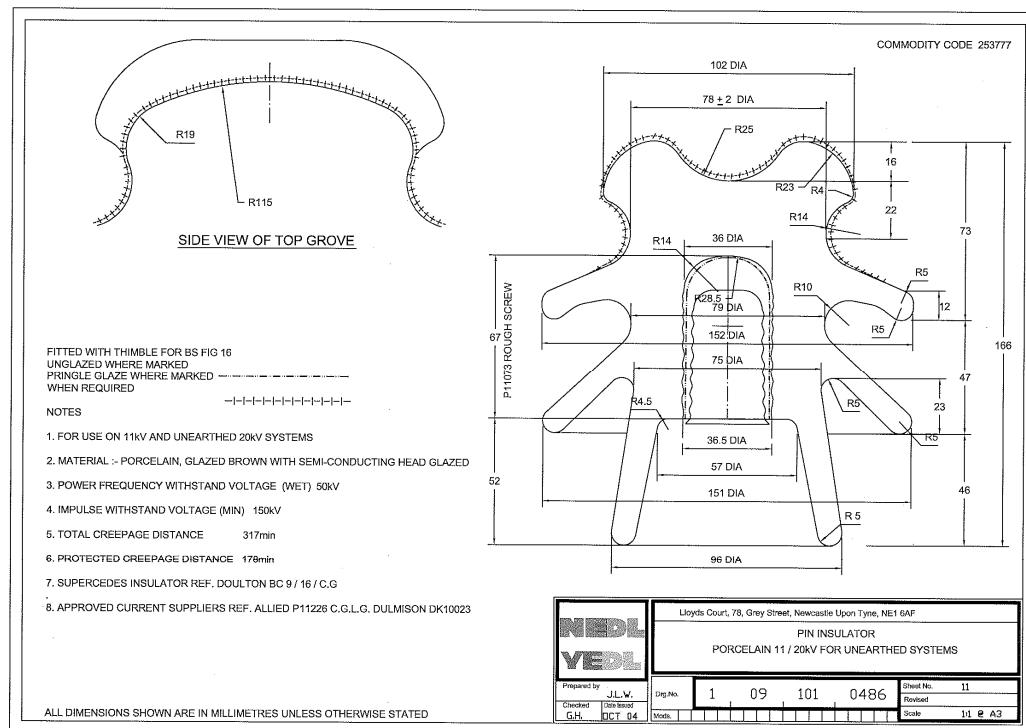
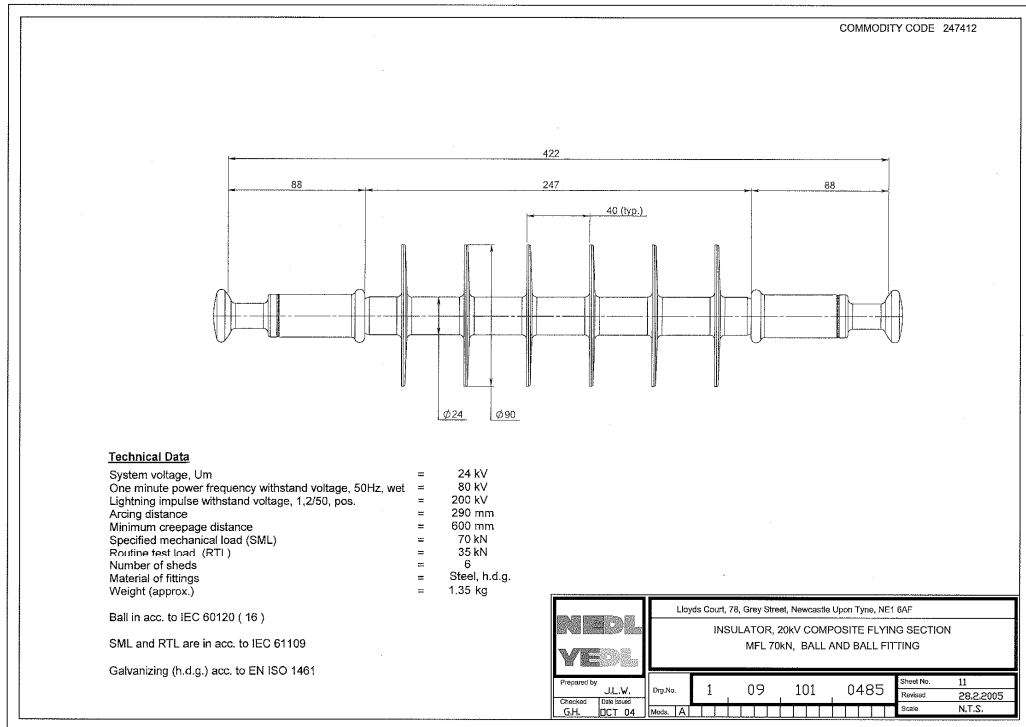
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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|-------------|---------|------------------------|------------------------|-----------|---|---------|------|-------------|--------|----------|--------|--|--|--|------|-----|-------|--------|
| N.E.D.L. CAT No 253616                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                            |             |         |                        |                        |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
|                                                                                                                                                                                                                                                                                                                                                                       |             |                            |             |         |                        |                        |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
| <p>NOTE</p> <p>1. FOR USE IN SUSPENSION INSULATOR SETS ON NORMAL 66kV &amp; 132kV OH LINES<br/>APPROVED TYPES TO BS 137 PART 2 (1973) :- DIM 'X' = 140mm<br/>PORCELAIN, ALLIED INSULATORS F80 / 140 / 2484A</p>                                                                                                                                                                                                                                          |             |                            |             |         |                        |                        |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
| Lloyds Court, 78, Grey Street, Newcastle Upon Tyne NE1 6AF                                                                                                                                                                                                                                                                                                                                                                                               |             |                            |             |         |                        |                        |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
| STRING INSULATOR UNIT<br>PORCELAIN (MFL 70kN)                                                                                                                                                                                                                                                                                                                                                                                                            |             |                            |             |         |                        |                        |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
| <table border="1"><tr><td rowspan="2" style="text-align: center; vertical-align: middle;"><b>NEDL</b><br/><b>VEDL</b></td><td>Prepared By</td><td>J.L.W.</td><td>Drg.No.</td><td>1    09    101    0485</td><td>Sheet No.</td><td>4</td></tr><tr><td>Checked</td><td>G.H.</td><td>Date Issued</td><td>APR 79</td><td>Revised.</td><td>2.3.05</td></tr><tr><td></td><td></td><td></td><td>Mod.</td><td>H J</td><td>Scale</td><td>N.T.S.</td></tr></table> |             | <b>NEDL</b><br><b>VEDL</b> | Prepared By | J.L.W.  | Drg.No.                | 1    09    101    0485 | Sheet No. | 4 | Checked | G.H. | Date Issued | APR 79 | Revised. | 2.3.05 |  |  |  | Mod. | H J | Scale | N.T.S. |
| <b>NEDL</b><br><b>VEDL</b>                                                                                                                                                                                                                                                                                                                                                                                                                               | Prepared By |                            | J.L.W.      | Drg.No. | 1    09    101    0485 | Sheet No.              | 4         |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Checked     | G.H.                       | Date Issued | APR 79  | Revised.               | 2.3.05                 |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                            | Mod.        | H J     | Scale                  | N.T.S.                 |           |   |         |      |             |        |          |        |  |  |  |      |     |       |        |

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|--------------------|----------------|-----------------|----------|------|----------|
| Document reference | NPS/ 001 / 006 |                 |          |      |          |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 27 of 49 |



CAUTION! - This document may be out of date if printed

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Feb - 10 | <b>Page</b> | 28 | <b>of</b> | 49 |

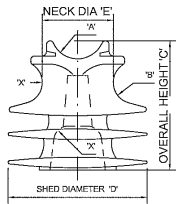


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|--------------------|----------------|-----------------|----------|---------------|
| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 29 of 49 |

CATALOGUE No 253701

HEAD DESIGNED FOR USE WITH BOTH HAND WRAPPED AND PRE - FORMED CONDUCTOR TIES. COMPATIBLE WITH POLYTHENE COVERED CONDUCTORS - NO STRIPPING REQUIRED. ALSO USABLE WITH BARE CONDUCTORS

SUITABLE FOR BS3288 STANDARD PINS



HIGHLY RESISTANT TO IMPACT DAMAGE, BREAKAGE AND VANDALISM

| PRODUCT CODE | SYSTEM VOLTAGE | ORDERING UNIT | WEIGHT (kg) | 'A' mm | 'B' mm | 'C' mm | 'D' mm | 'E' mm |
|--------------|----------------|---------------|-------------|--------|--------|--------|--------|--------|
| 56431-04     | 17 / 25kV      | EACH          | 0.90        | 25     | 22     | 140    | 191    | 73     |

| ELECTRICAL DATA               |                        |
|-------------------------------|------------------------|
| TEST                          | 56431-04 SPECIFICATION |
| CREEPAGE (mm) (X-X)           | 349                    |
| TAUT STRING (mm)              | 190                    |
| IMPULSE FLASHOVER(DRY)(kVp)   | 150                    |
| IMPULSE WITHSTAND (DRY) (kVp) | 137                    |
| AC 1min FLASHOVER(DRY)(kV)    | 94                     |
| AC 1min WITHSTAND (DRY)(kV)   | 86                     |
| AC 1min FLASHOVER (WET)(kV)   | 52                     |
| AC 1 min WITHSTAND (WET)(kV)  | 48                     |

| MECHANICAL DATA            |                        |
|----------------------------|------------------------|
| TEST                       | 56431-04 SPECIFICATION |
| CANTILEVER LOAD            | 13                     |
| RADIO INTERFERENCE VOLTAGE | 0 at 15kV              |

1. SPECIFIED ELECTRICAL TESTS

1.1 DRY LIGHTNING IMPULSE TESTS : IEC 383 / IEC 1109

1.2 POWER FREQUENCY VOLTAGE TEST : IEC 383 / IEC 1109

1.3 TRACKING RESISTANCE TEST : ASTM D 2303

1.4 IMPACT RESISTANCE TEST : VARIOUS

2. SPECIFIED MECHANICAL TESTS

2.1 CANTILEVER TEST : ANSI C29.1

3. SPECIFIED ENVIRONMENTAL TESTS

3.1 1000hr. SALT / FOG TEST : IEC 1109

3.2 ULTRAVIOLET EXPOSURE TEST : ASTM G26

3.3 20 MONTH SEVERE SERVICE TEST : ANSI C-2.1



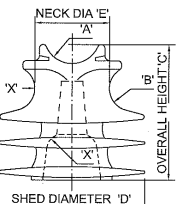
Prepared by J.L.W.  
Checked G.H. MAR 05

|                                                             |              |
|-------------------------------------------------------------|--------------|
| Lloyds Court, 78, Grey Street, Newcastle Upon Tyne, NE1 6AF |              |
| 11KV COMPOSITE ANTI - VANDAL PIN INSULATOR                  |              |
| Dwg No. 1 09 101 0486                                       | Sheet No. 12 |
| Scale                                                       | N.T.S.       |

CATALOGUE No 253833

HEAD DESIGNED FOR USE WITH BOTH HAND WRAPPED AND PRE - FORMED CONDUCTOR TIES. COMPATIBLE WITH POLYTHENE COVERED CONDUCTORS - NO STRIPPING REQUIRED. ALSO USABLE WITH BARE CONDUCTORS

SUITABLE FOR BS3288 STANDARD PINS



HIGHLY RESISTANT TO IMPACT DAMAGE, BREAKAGE AND VANDALISM

| PRODUCT CODE | SYSTEM VOLTAGE | ORDERING UNIT | WEIGHT (kg) | 'A' mm | 'B' mm | 'C' mm | 'D' mm | 'E' mm |
|--------------|----------------|---------------|-------------|--------|--------|--------|--------|--------|
| 56431-07     | 30 / 36 kV     | EACH          | 1.17        | 25     | 22     | 181    | 191    | 73     |

| ELECTRICAL DATA               |                        |
|-------------------------------|------------------------|
| TEST                          | 56431-07 SPECIFICATION |
| CREEPAGE (mm) (X-X)           | 533                    |
| TAUT STRING (mm)              | 241                    |
| IMPULSE FLASHOVER(DRY)(kVp)   | 184                    |
| IMPULSE WITHSTAND (DRY) (kVp) | 180                    |
| AC 1min FLASHOVER(DRY)(kV)    | 110                    |
| AC 1min WITHSTAND (DRY)(kV)   | 105                    |
| AC 1min FLASHOVER (WET)(kV)   | 66                     |
| AC 1 min WITHSTAND (WET)(kV)  | 55                     |

| MECHANICAL DATA            |                        |
|----------------------------|------------------------|
| TEST                       | 56431-07 SPECIFICATION |
| CANTILEVER LOAD            | 13                     |
| RADIO INTERFERENCE VOLTAGE | 0 at 15kV              |

1. SPECIFIED ELECTRICAL TESTS

1.1 DRY LIGHTNING IMPULSE TESTS : IEC 383 / IEC 1109

1.2 POWER FREQUENCY VOLTAGE TEST : IEC 383 / IEC 1109

1.3 TRACKING RESISTANCE TEST : ASTM D 2303

1.4 IMPACT RESISTANCE TEST : VARIOUS

2. SPECIFIED MECHANICAL TESTS

2.1 CANTILEVER TEST : ANSI C29.1

3. SPECIFIED ENVIRONMENTAL TESTS

3.1 1000hr. SALT / FOG TEST : IEC 1109

3.2 ULTRAVIOLET EXPOSURE TEST : ASTM G26

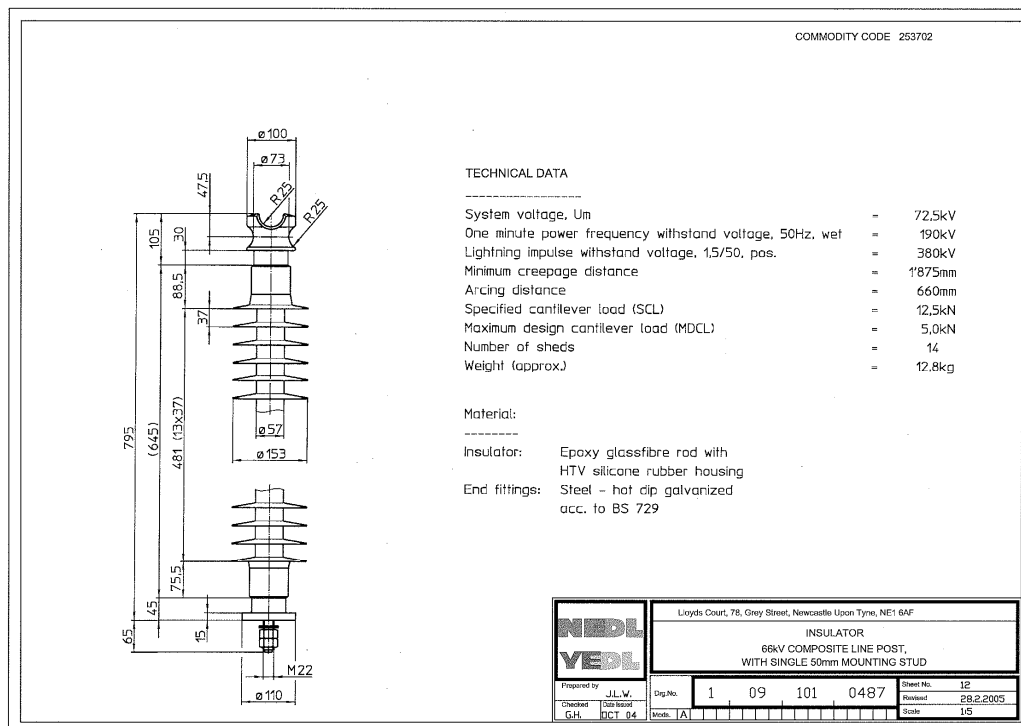
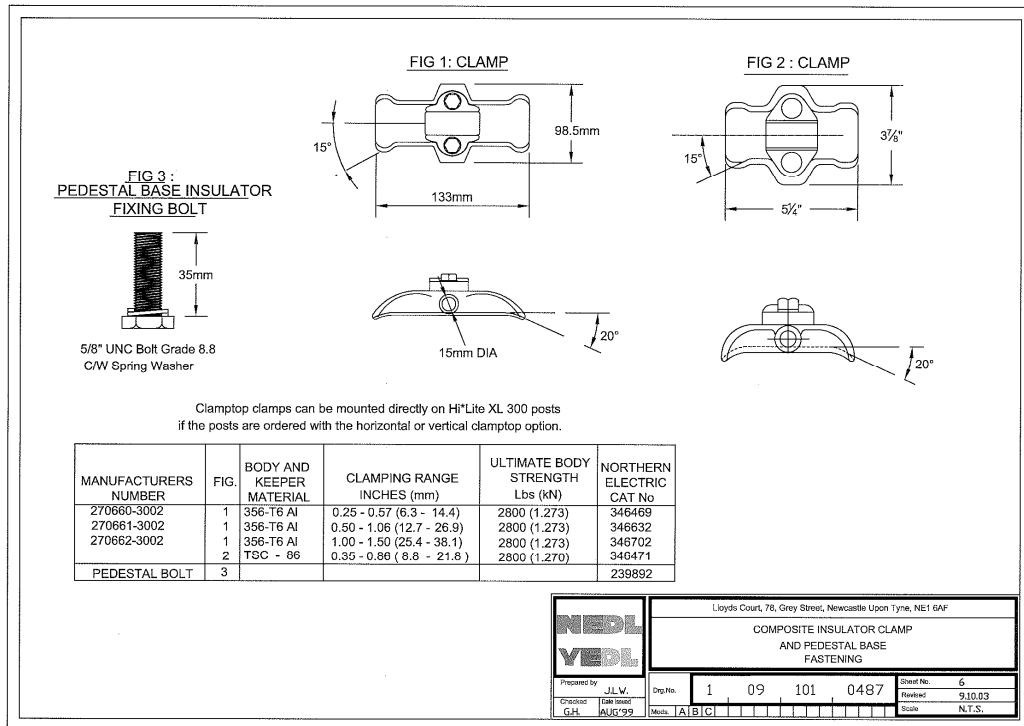
3.3 20 MONTH SEVERE SERVICE TEST : ANSI C-2.1



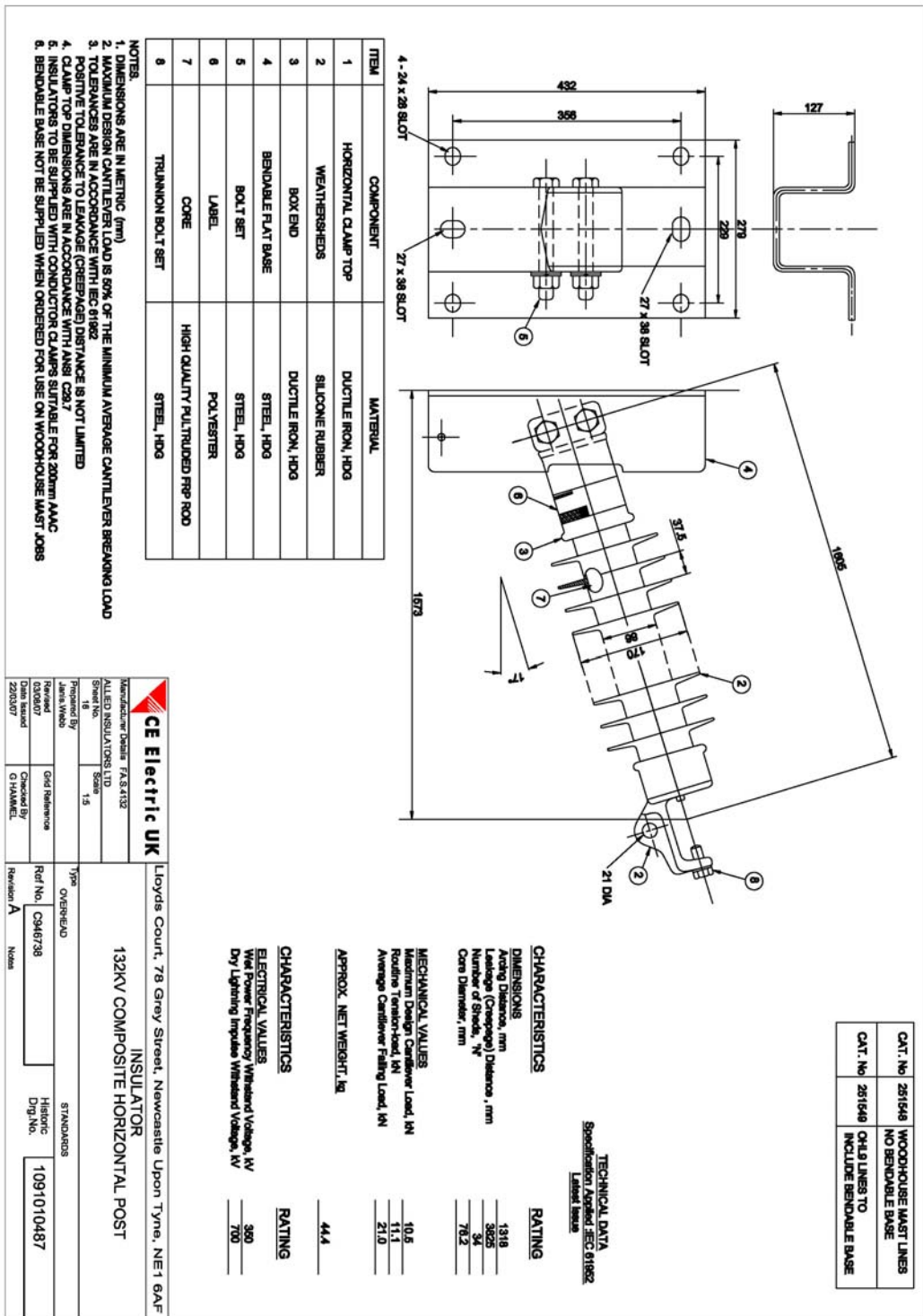
Prepared by J.L.W.  
Checked G.H. MAR 05

|                                                                                           |              |
|-------------------------------------------------------------------------------------------|--------------|
| Lloyds Court, 78, Grey Street, Newcastle Upon Tyne, NE1 6AF                               |              |
| 20KV COMPOSITE PIN INSULATOR SUITABLE FOR 20KV EARTHED APPLICATIONS INCLUDING ANTI-VANDAL |              |
| Dwg No. 1 09 101 0486                                                                     | Sheet No. 13 |
| Scale                                                                                     | N.T.S.       |

|                    |                |                 |          |               |
|--------------------|----------------|-----------------|----------|---------------|
| Document reference | NPS/ 001 / 006 |                 |          |               |
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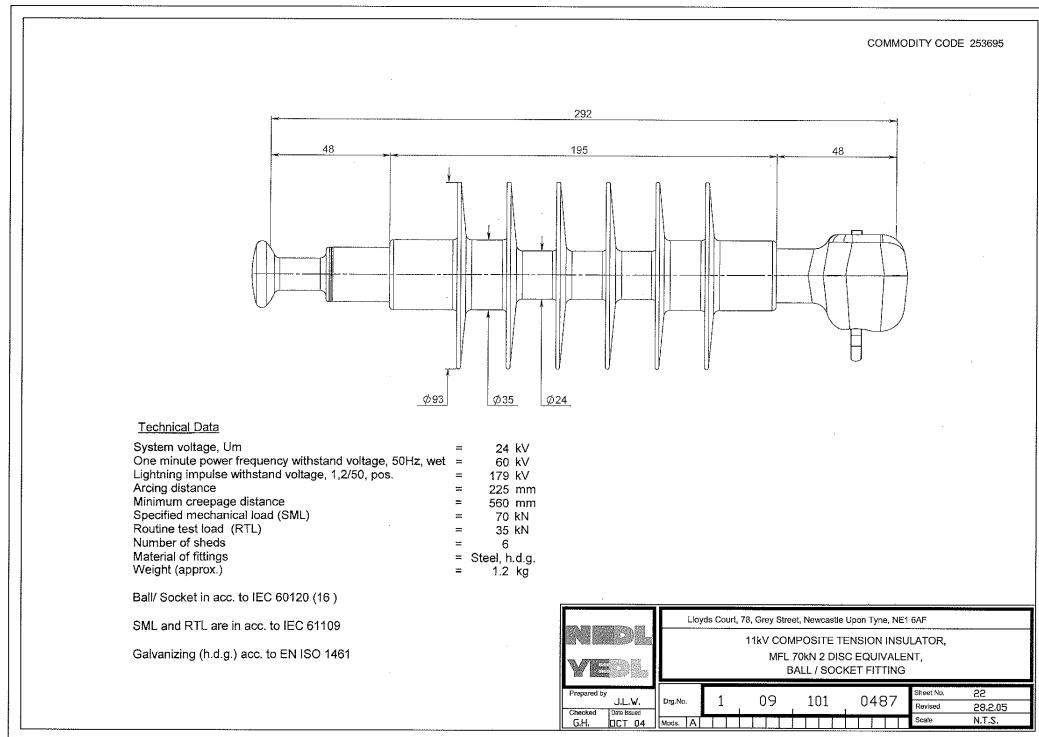
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| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 31 of 49 |



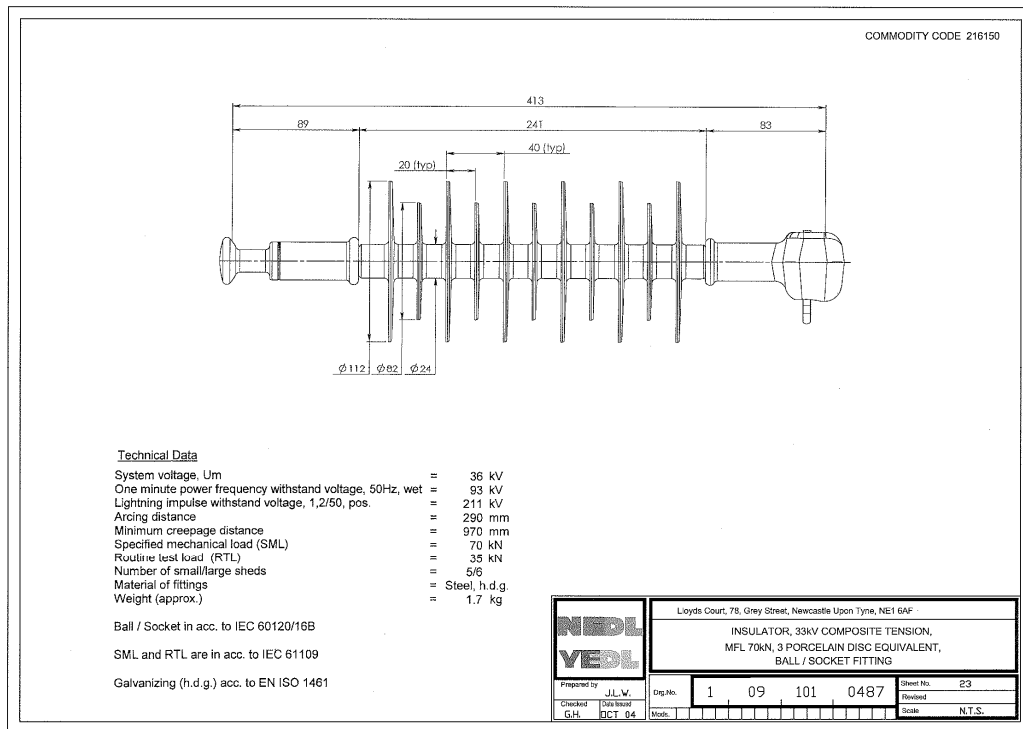
**TECHNICAL DATA**  
Specification Applied : IEC 61952  
Latest Issue

CAT. No 251551 SUPPLIED WITHOUT STOOL.  
CAT. No 251550 SUPPLIED WITH SUPPORT STOOL  
AND ARMOUR GRIP CLAMPS

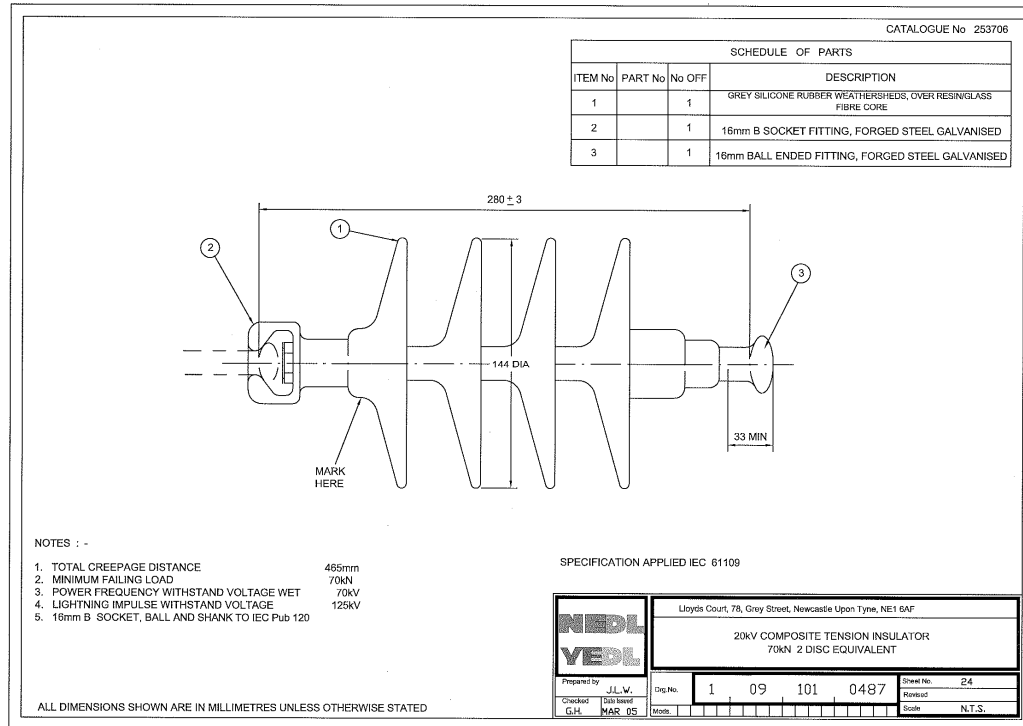
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| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 33 | of | 49 |



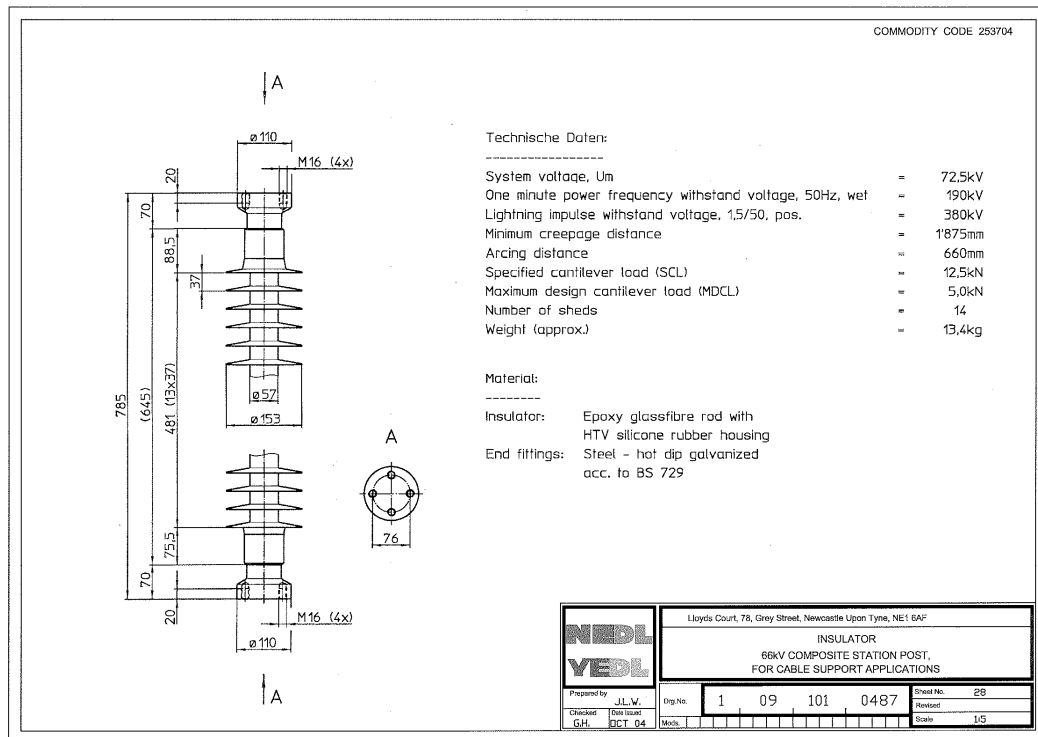
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| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 34 | of | 49 |



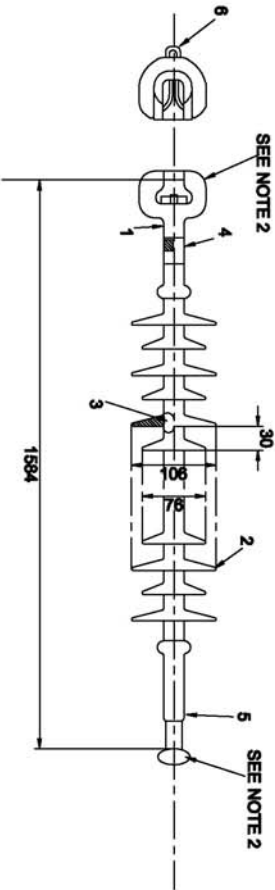
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| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 35 | of | 49 |



|                    |                |                 |          |      |    |    |    |
|--------------------|----------------|-----------------|----------|------|----|----|----|
| Document reference | NPS/ 001 / 006 |                 |          |      |    |    |    |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page | 36 | of | 49 |



|                    |                |                 |          |               |
|--------------------|----------------|-----------------|----------|---------------|
| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 37 of 49 |



| ITEM | COMPONENTS     | MATERIAL                       | REQD. |
|------|----------------|--------------------------------|-------|
| 1    | SOCKET FITTING | HIGH GRADE F.S. OR D.I. HDG    | 1     |
| 2    | WEATHERSHEDS   | SILICONE RUBBER                | 1     |
| 3    | CORE           | HIGH QUALITY PULTRUDED FRP ROD | 1     |
| 4    | LABEL          | POLYESTER                      | 1     |
| 5    | BALL FITTING   | HIGH GRADE FORGED STEEL, HDG   | 1     |
| 6    | SPLIT PIN      | STAINLESS STEEL                | 1     |

TECHNICAL DATA  
SPECIFICATION APPLIED : IEC61109, LATEST ISSUE  
CHARACTERISTICS

RATING

DIMENSIONS

ARcing DISTANCE mm  
LEAKAGE (CREEPAGE) DISTANCE mm  
NUMBER OF SHEDS 'N'  
CORE DIAMETER mm

1348  
3702  
43  
17

MECHANICAL VALUES

SPECIFIED MECHANICAL LOAD IN  
ROUTINE TENSION LOAD IN

125  
63

APPROX. NET WEIGHT. kg

6.0

ELECTRICAL VALUES

WET POWER FREQUENCY WITHSTAND VOLTAGE kV  
DRY LIGHTNING IMPULSE WITHSTAND VOLTAGE kV

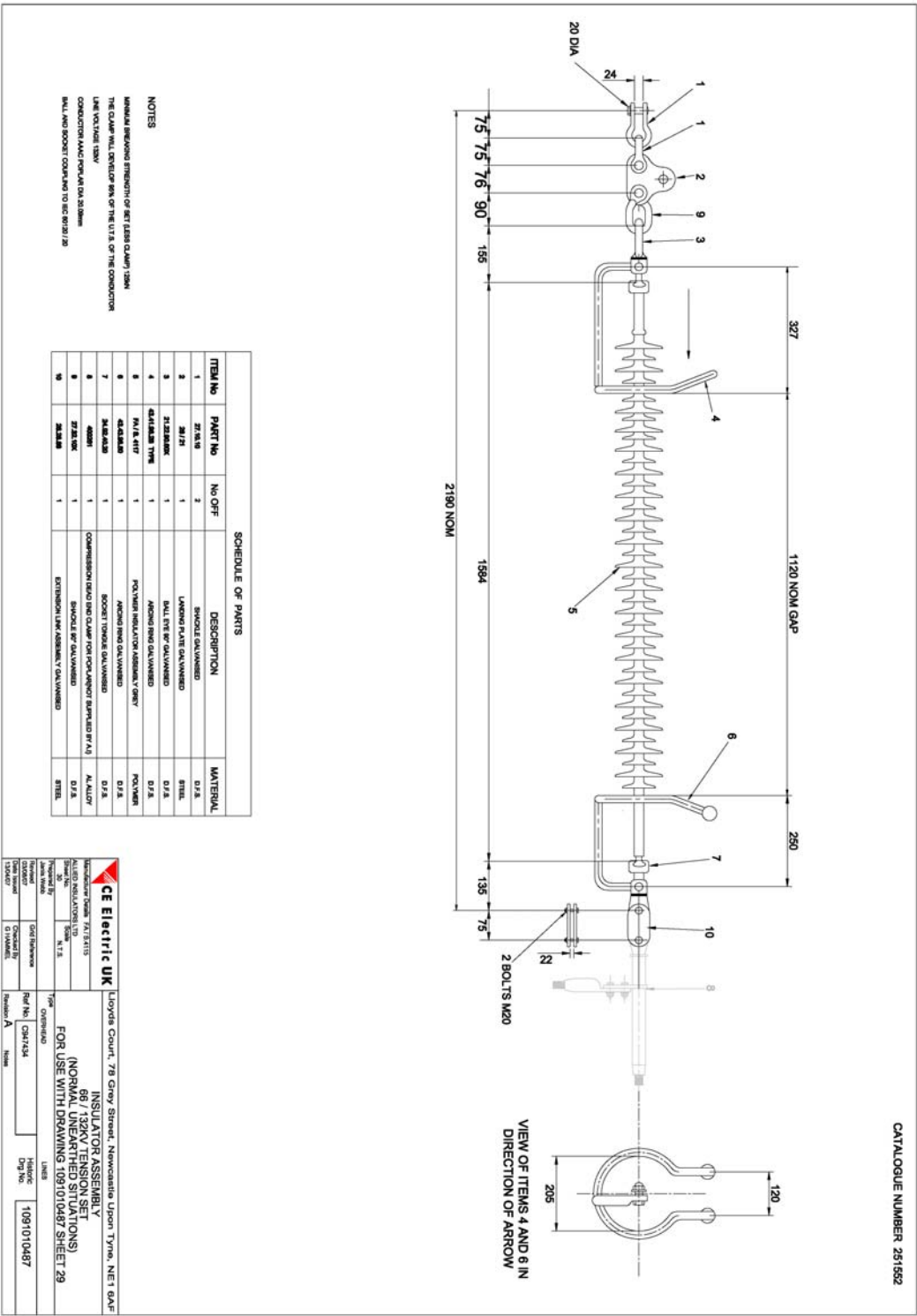
425  
765

NOTES

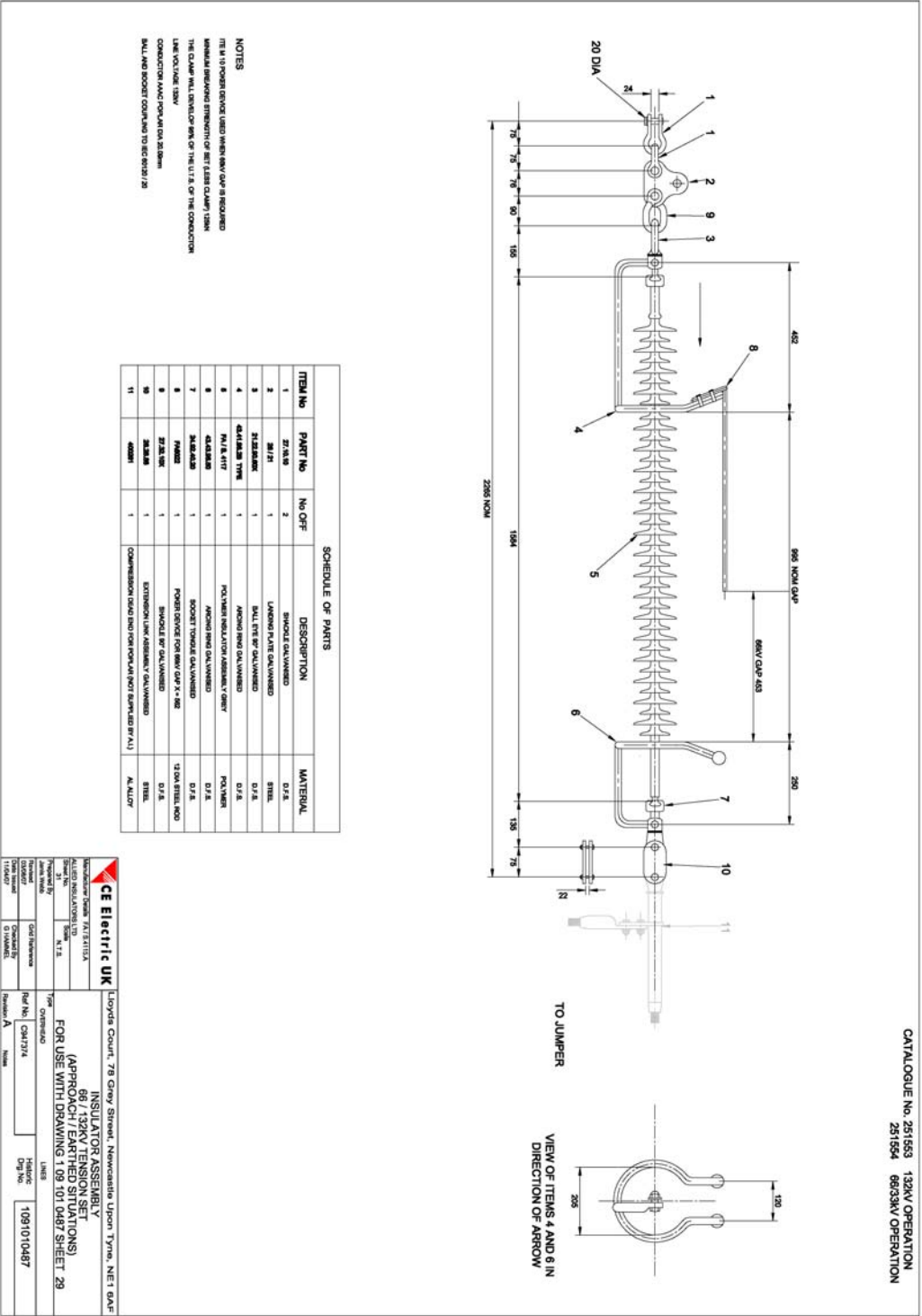
1. DIMENSIONS ARE IN MILLIMETRES
2. SOCKET AND BALL COUPLINGS CONFORM TO GAUGES FOR IEC PUB60120 / 20
3. TOLERANCES ARE IN ACCORDANCE WITH IEC 61109, LATEST ISSUE.  
POSITIVE TOLERANCE TO LEAKAGE (CREEPAGE) DISTANCE IS NOT LIMITED

|                                                                                                           |           |                                                                                                              |            |
|-----------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|------------|
|  <b>CE Electric UK</b> |           | Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF                                                   |            |
| Manufacturer Details FAS.4117                                                                             |           | <b>INSULATOR</b><br><b>132KV COMPOSITE TENSION / STAY</b><br><b>(120KN WITH 20MM BALL / SOCKET COUPLING)</b> |            |
| ALLIED INSULATORS LTD                                                                                     |           |                                                                                                              |            |
| Street No.                                                                                                | Scale 1:5 |                                                                                                              |            |
| 29                                                                                                        |           |                                                                                                              |            |
| Prepared By                                                                                               |           | TYPE OVERHEAD                                                                                                |            |
| Checked By                                                                                                |           | Ref No.                                                                                                      | C947326    |
| Drawn By                                                                                                  |           | Historic                                                                                                     | 1091010487 |
| 11/04/07                                                                                                  |           | Checked By                                                                                                   | G HAMMEL   |
|                                                                                                           |           | Revision                                                                                                     | Notes      |

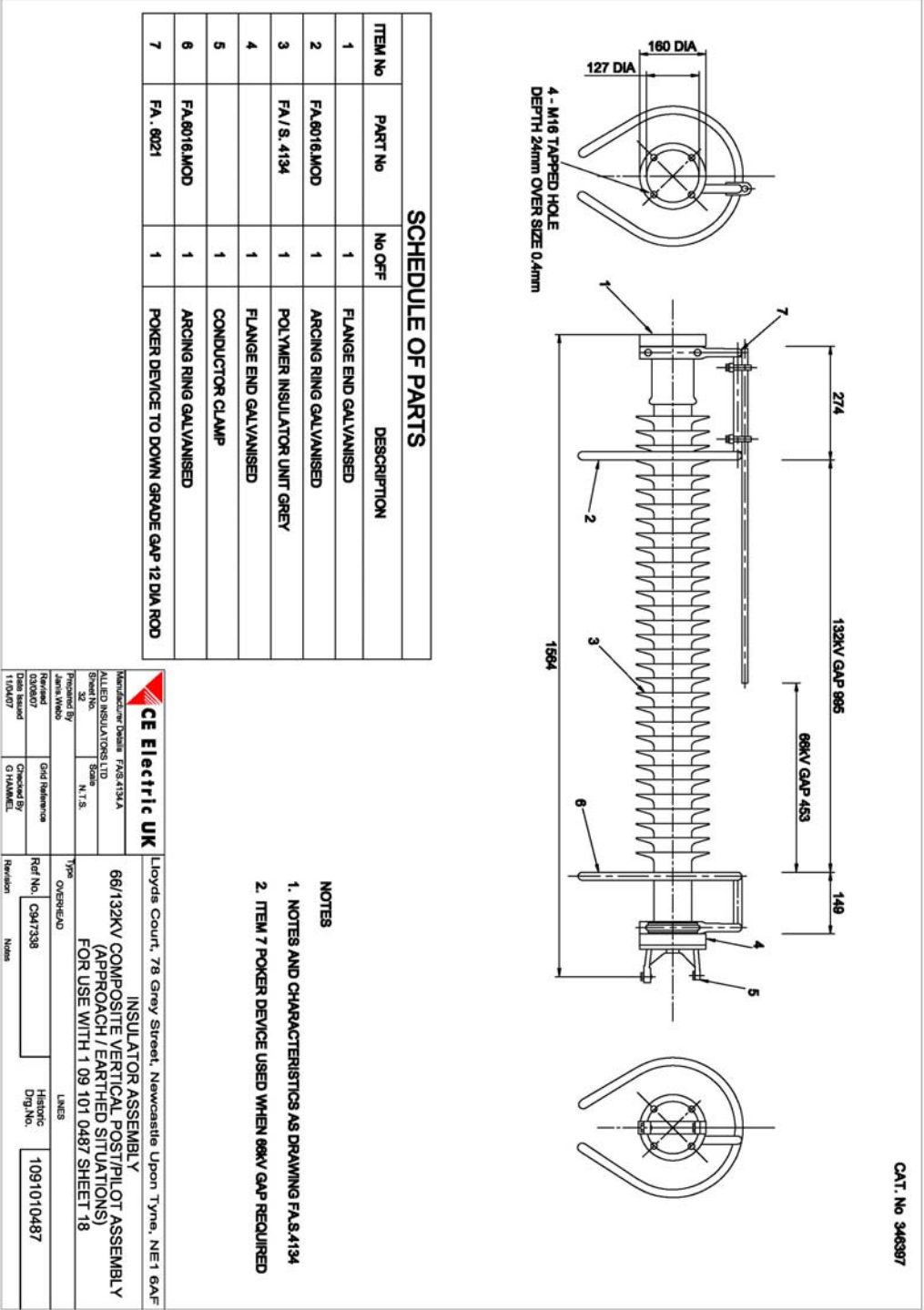
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|--------------------|----------------|-----------------|----------|---------------|
| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 38 of 49 |



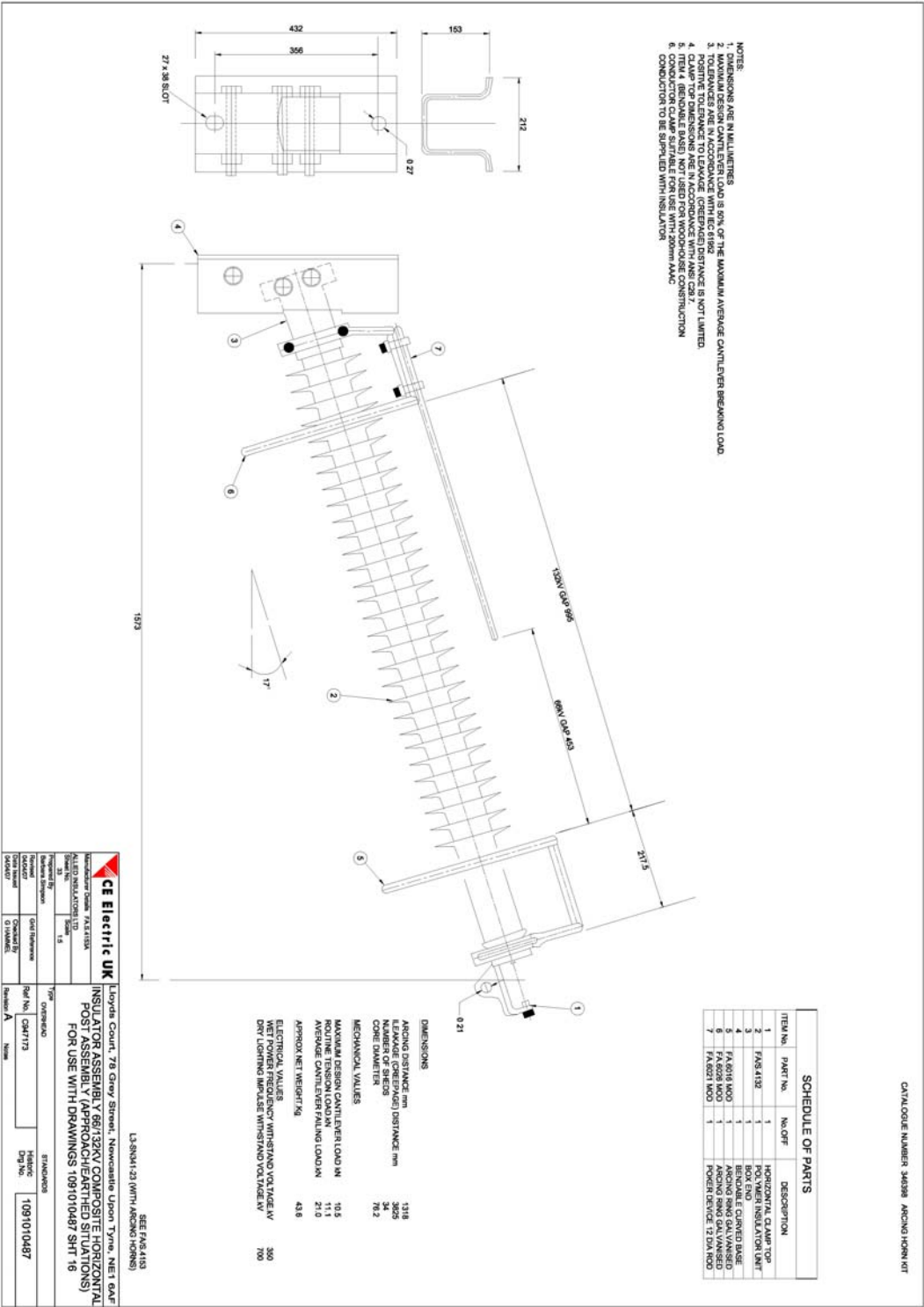
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| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 39 of 49 |



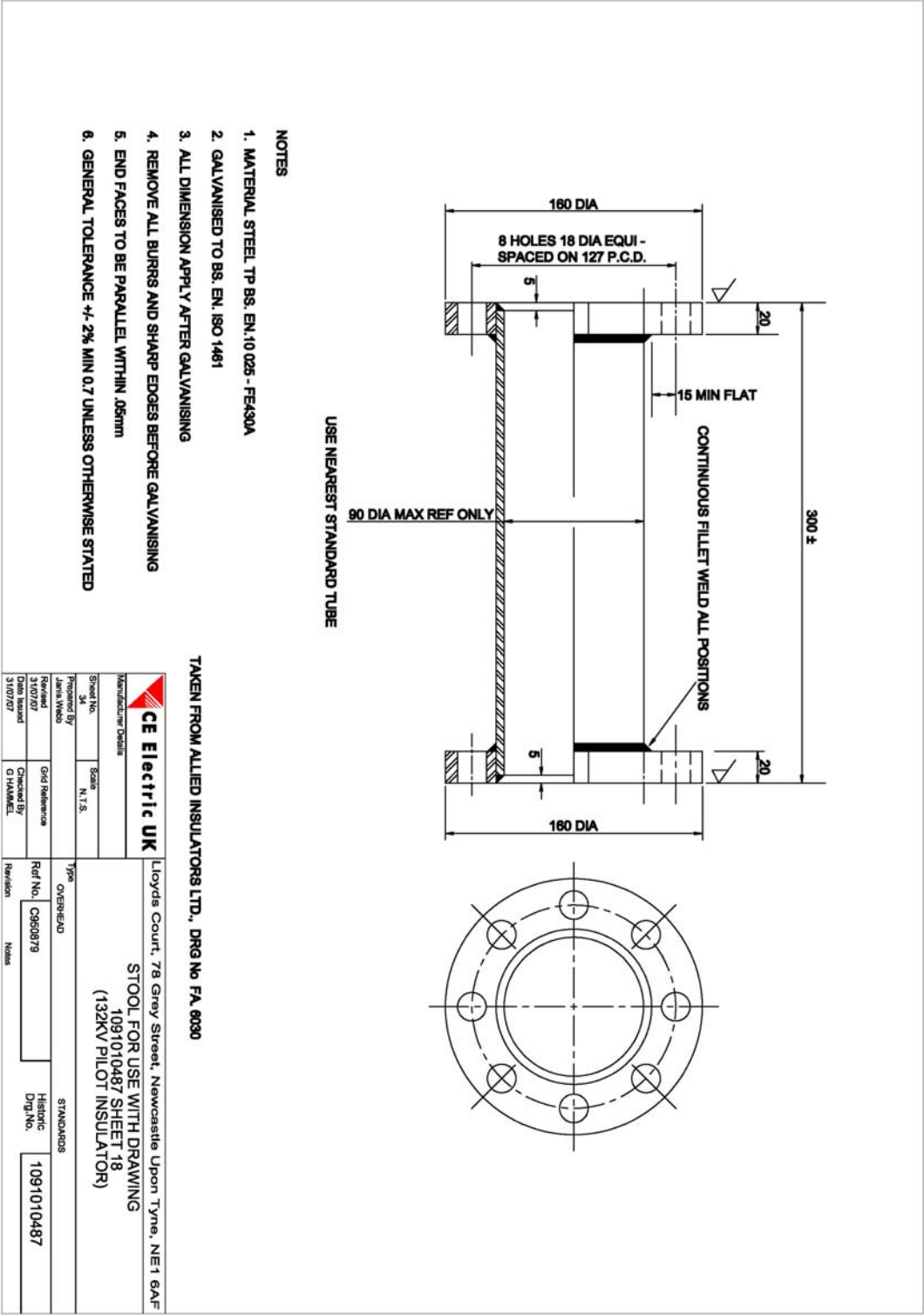
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| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 40 of 49 |



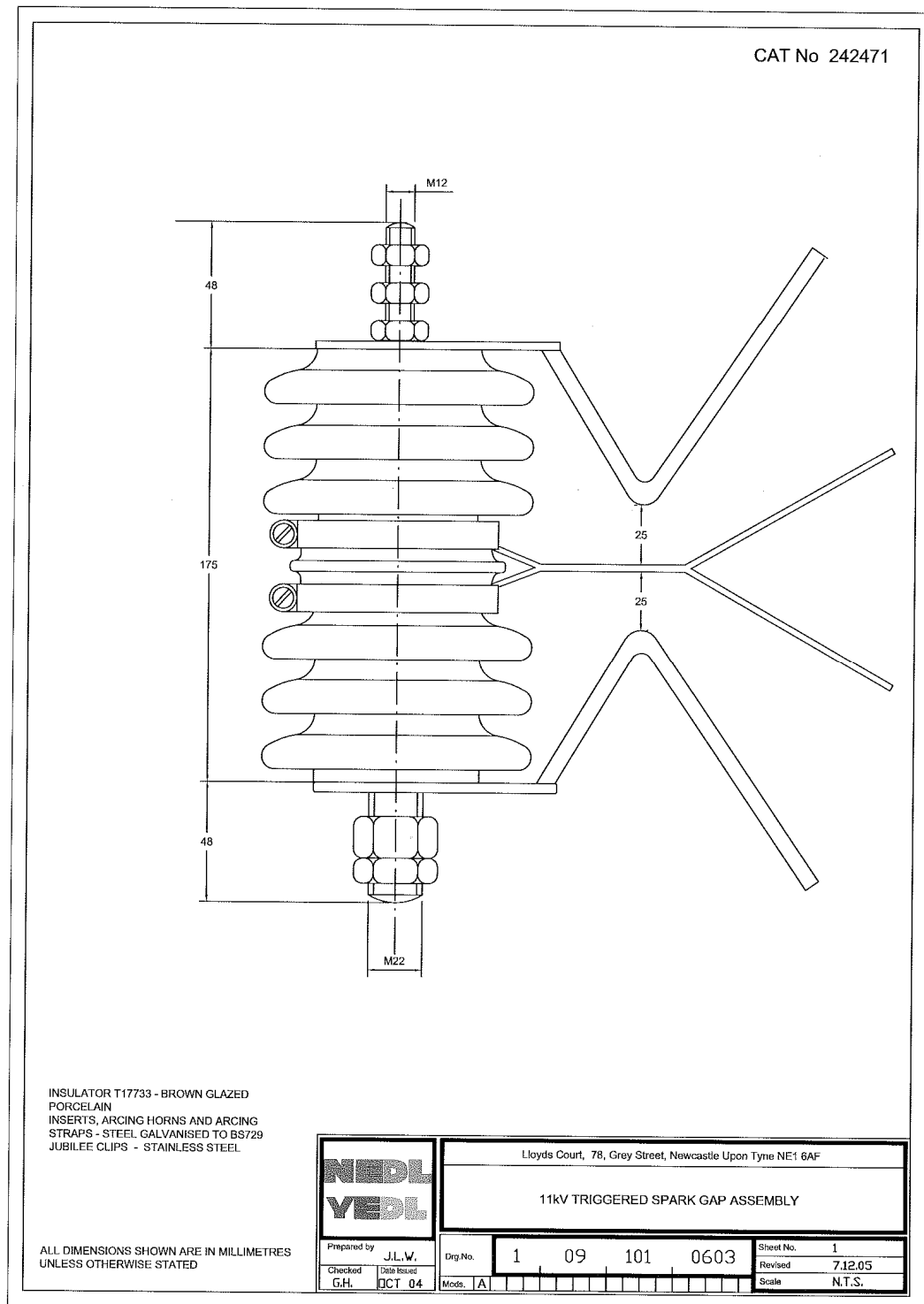
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| Document reference | NPS/ 001 / 006 |                 |          |               |
| Version:-          | 2.0            | Date of Issue:- | Feb - 10 | Page 41 of 49 |



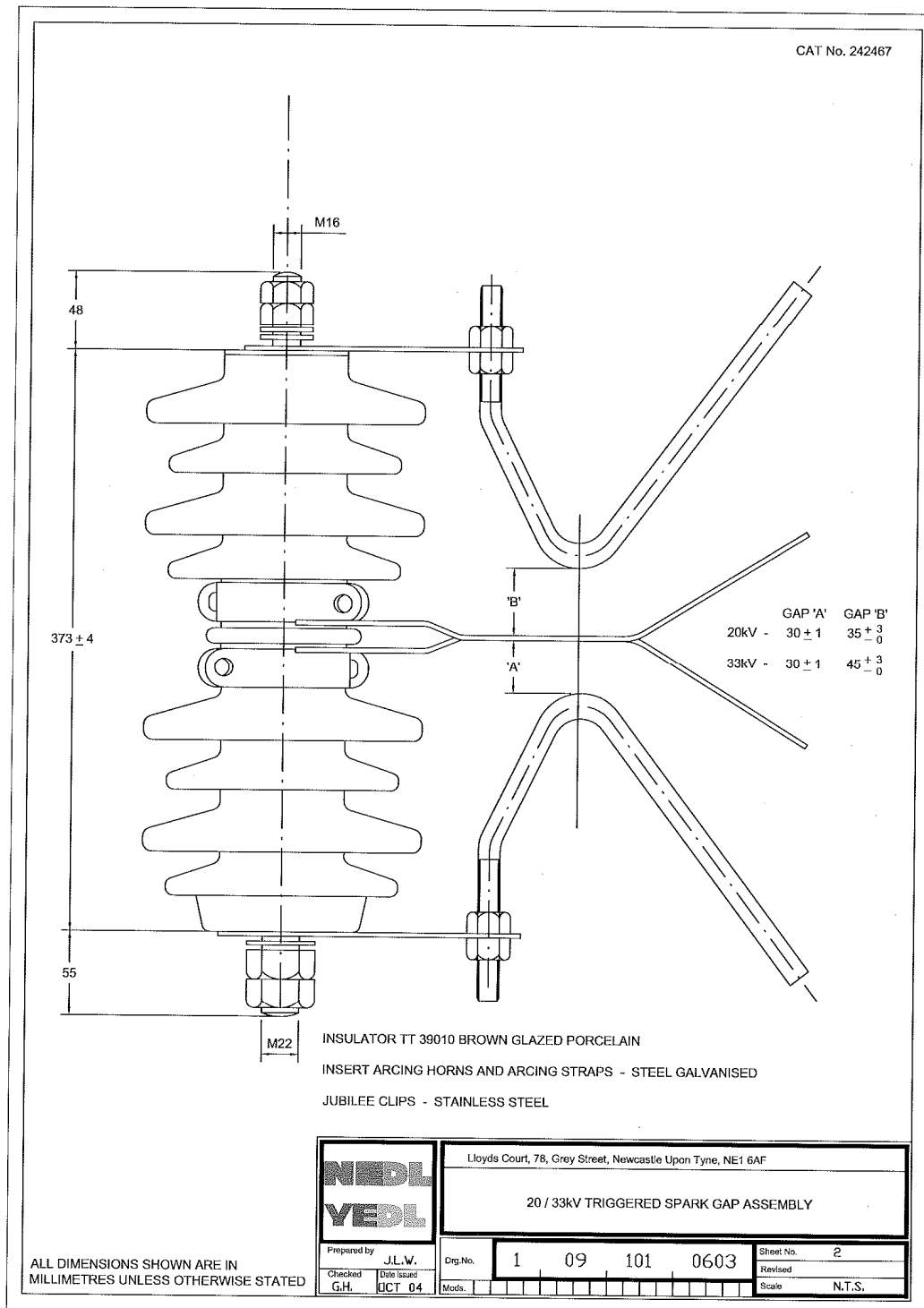
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|--------------------|----------------|-----------------|----------|---------------|
| Document reference | NPS/ 001 / 006 |                 |          |               |
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| Document reference | NPS/ 001 / 006 |                 |          |      |    |    |    |
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|                    |                |                 |          |      |    |    |    |
|--------------------|----------------|-----------------|----------|------|----|----|----|
| Document reference | NPS/ 001 / 006 |                 |          |      |    |    |    |
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|                           |     |                        |          |             |    |           |    |
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| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
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## APPENDIX 3 - Self Certification Conformance Declaration

Line insulators manufactured in accordance with ENA TS 43-93 shall comply with the latest issues of the relevant international and British Standards.

ENA TS 43-93 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENA TS 43-93 and the clauses of the aforementioned Standards relevant to line insulators for use on CE Electric UK designs of overhead lines.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes. Additionally manufacturers shall provide test evidence for all completed tests.

### Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

### Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENA TS' as appropriate

**Manufacturer:**

**Product Reference:**

**Name:**

**Signature:**

**Date:**

Appendix 3 Tables 1 to 7 follow:

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
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**Table 1**

| <b>EN 60383- 1</b>  |                                                                   |                  | <b>ENA TS 43– 93</b> |                                    |                  |         |
|---------------------|-------------------------------------------------------------------|------------------|----------------------|------------------------------------|------------------|---------|
| Clause / Sub-clause | Requirement                                                       | Conformance Code | Clause / Sub-clause  | Requirement                        | Conformance Code | Remarks |
| 5                   | Identification of insulators                                      |                  | 4.1                  |                                    |                  |         |
| Section 4           | Test procedures for electrical tests                              |                  | 6.2                  | Design and Type Tests <sup>2</sup> |                  |         |
| Section 5           | Test procedures for mechanical and other tests <sup>1</sup>       |                  | 6.3                  | Sample Tests                       |                  |         |
|                     |                                                                   |                  | 6.4                  | Routine Tests                      |                  |         |
| 18, 19              | Electromechanical failing load test, Mechanical failing load test |                  | 8.1 or 8.2           | Mechanical Performance             |                  |         |

**Table 2**

| <b>EN 60383- 1</b>  |                                   |                  | <b>ENA TS 43– 93</b> |             |                  |         |
|---------------------|-----------------------------------|------------------|----------------------|-------------|------------------|---------|
| Clause / Sub-clause | Requirement                       | Conformance Code | Clause / Sub-clause  | Requirement | Conformance Code | Remarks |
| 9                   | Lightning Impulse Voltage tests   |                  | 6.1                  | General     |                  |         |
| 10                  | Wet power frequency voltage tests |                  |                      |             |                  |         |

<sup>1</sup> With the exception of Clauses 18 and 19 of EN 60383 which is called by Clause 8.2 of ENA TS 43- 93.

<sup>2</sup> The selection of the appropriate type, sample and routine tests within section 4 and 5 of EN 60383- 1 should be based on Appendix A of ENA TS 43- 93

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
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**Table 3**

| <b>BS EN 61466-1</b> |             |                          | <b>ENA TS 43- 93</b> |                      |                          |         |
|----------------------|-------------|--------------------------|----------------------|----------------------|--------------------------|---------|
| Clause / Sub-clause  | Requirement | Confor-<br>mance<br>Code | Clause / Sub-clause  | Requirement          | Confor-<br>mance<br>Code | Remarks |
| 6                    | Marking     |                          | 4.1                  | Marking <sup>3</sup> |                          |         |
|                      |             |                          | 4.2, 4.3, 4.4        | Materials            |                          |         |

**Table 4**

| <b>IEC 61952</b>    |                                                                                 |                          | <b>ENA TS 43- 93</b> |                                         |                          |         |
|---------------------|---------------------------------------------------------------------------------|--------------------------|----------------------|-----------------------------------------|--------------------------|---------|
| Clause / Sub-clause | Requirement                                                                     | Confor-<br>mance<br>Code | Clause / Sub-clause  | Requirement                             | Confor-<br>mance<br>Code | Remarks |
| 7.2                 | Electrical Tests                                                                |                          | 6.1                  | General                                 |                          |         |
| 6                   | Design Tests                                                                    |                          | 6.2                  | Design and Type Test                    |                          |         |
| 7                   | Type Tests <sup>4,5</sup>                                                       |                          | 6.2                  | Design and Type Test                    |                          |         |
| 8                   | Sample Tests                                                                    |                          | 6.3                  | Sample Tests                            |                          |         |
| 9                   | Routine Tests                                                                   |                          | 6.4                  | Routine Tests                           |                          |         |
| Annex C             | Ageing test under operating voltage, simulating weather conditions <sup>6</sup> |                          | 6.5                  | Tracking and Erosion Tests              |                          |         |
| 7.3                 | Mechanical Tests                                                                |                          | 8.2                  | Mechanical Performance (Composite Post) |                          |         |

<sup>3</sup> The remainder of Clauses 4 and 5 of ENA TS 43- 93 relate to the specific design requirements and characteristics of insulators commonly used in the UK.

<sup>4</sup> With the exception of Clause 7.3 of IEC 61952 which is called by Clause 8.2 of ENA TS 43- 93.

<sup>5</sup> With the exception of Clause 7.2 of IEC 61952 which is called by Clause 6.1 of ENA TS 43- 93.

<sup>6</sup> Tests defined within Annex C are not called by Clause 6.4.3 of IEC 61952 but are discussed in the introduction to the Standard. The tests called in Annex C of IEC 61109 may therefore be considered appropriate for composite post insulators used in heavy or very heavy pollution environments as defined in IEC 60815.

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
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**Table 5**

| <b>IEC 61109</b>    |                                                                                                       |                  | <b>ENA TS 43- 93</b> |                                           |                  |         |
|---------------------|-------------------------------------------------------------------------------------------------------|------------------|----------------------|-------------------------------------------|------------------|---------|
| Clause / Sub-clause | Requirement                                                                                           | Conformance Code | Clause / Sub-clause  | Requirement                               | Conformance Code | Remarks |
| 6.1, 6.2            | Dry lightning impulse withstand voltage test <sup>9</sup> , Wet power frequency tests Both to IEC 383 |                  | 6.1                  | General                                   |                  |         |
| 5                   | Design Tests                                                                                          |                  | 6.2                  | Design and Type Test                      |                  |         |
| 6                   | Type Test <sup>7 8</sup>                                                                              |                  | 6.2                  | Design and Type Test                      |                  |         |
| 7                   | Sampling Test                                                                                         |                  | 6.3                  | Sample Tests                              |                  |         |
| 8                   | Routine Test                                                                                          |                  | 6.4                  | Routine Tests                             |                  |         |
| Annex C             | Ageing test under operating voltage simulating weather conditions <sup>9</sup>                        |                  | 6.5                  | Tracking and Erosion Tests                |                  |         |
| 6.4                 | Mechanical load- time test                                                                            |                  | 8.2                  | Mechanical Performance (Composite String) |                  |         |

<sup>7</sup> With the exception of Clause 6.1 and 6.2 of IEC 61109 which are called by Clause 6.1 of ENA TS 43- 93.

<sup>8</sup> With the exception of Clause 6.4 of IEC 61109 which is called by Clause 8.2 of ENA TS 43- 93.

<sup>9</sup>Tests defined within Annex C are of IEC 61109 are appropriate for composite string insulators used in heavy or very heavy pollution environments as defined in IEC 60815.

|                           |     |                        |          |             |    |           |    |
|---------------------------|-----|------------------------|----------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NPS/ 001 / 006         |          |             |    |           |    |
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**Table 6**

| <b>IEC 60815</b>    |                                                                                       |                  | <b>ENA TS 43- 93</b> |             |                  |         |
|---------------------|---------------------------------------------------------------------------------------|------------------|----------------------|-------------|------------------|---------|
| Clause / Sub-clause | Requirement                                                                           | Conformance Code | Clause / Sub-clause  | Requirement | Conformance Code | Remarks |
| 1- 7                | Guide for the selection of insulators in respect of polluted conditions <sup>10</sup> |                  |                      | Pollution   |                  |         |

<sup>10</sup> ENA TS 43- 93 calls IEC 60815 for the selection of the appropriate specific creepage distance. IEC 60815 also describes general design criteria for porcelain and glass insulators which should be considered when selecting insulators for specific environments.

**Table 7**

| <b>IEC 60587-2007</b> |                                                                                                                                                                      |                  |                     |             |                  |         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-------------|------------------|---------|
| Clause / Sub-clause   | Requirement                                                                                                                                                          | Conformance Code | Clause / Sub-clause | Requirement | Conformance Code | Remarks |
|                       | Evaluation of electrical insulating materials for use under severe ambient conditions<br>Method 1: constant tacking method or<br>Method 2: stepwise tracking voltage |                  |                     |             |                  |         |